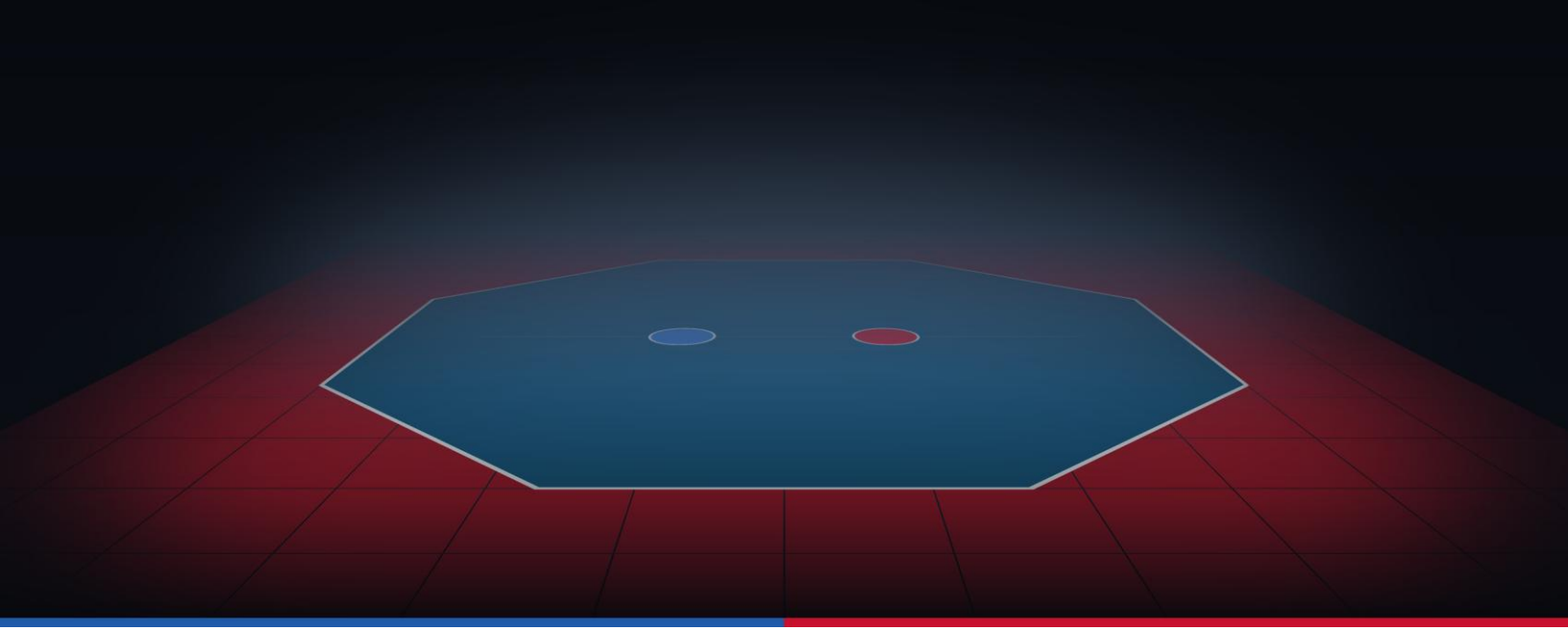


BCIT CAPSTONE · 2026–2027

Open Stance

겨루기 Gyeorugi · Game Design Document

Real taekwondo sparring, one on one: read the distance, choose your kick with a flick of the Skill Stick, and land it before your opponent does.

VERSION	0.12 draft · updated 2026-10-30
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1 Overview

Open Stance captures the rhythm and decision-making of real taekwondo: distance, timing, stance, guard, counters and scoring. It starts as a sparring game built around motion capture of real practitioners, and grows into the home of competitive taekwondo sparring: tournaments, team events, online competition and para taekwondo. Open Stance is only about sparring; forms (poomsae) and board breaking are out of scope.

Phases. Everything in this document is part of v1.0, the game this capstone delivers: the First Half builds the sparring core, the Second Half adds Career (1.14). A **[Future]** tag marks the exceptions: expansion planned for after v1.0. Nothing tagged **[Future]** is essential to the first v1.0 release. Those sections are documented so the design is complete and so there is a plan if the main features finish ahead of schedule; they are the first thing to leave out when time is short and they never block a milestone.

1.1 Working Title

Open Stance: Gyeorugi. In taekwondo sparring, an open stance is the position where the two fighters lead with opposite legs, so their trunks face each other and the roundhouse kick is open. Gyeorugi (겨루기) is the sport's own word for sparring. The title tells the player that the game is about stance, position and competitive sparring, presented in taekwondo's own terms.

1.2 Current Status

Status on 2026-09-28: pre-production, in week 4 (prototyping toward Milestone 1) of the First Half. GDD Draft 1 was presented today, to the instructor and second reader, with feedback due 2 November. The design is a draft that has not yet been reviewed, and no gameplay prototype exists. The character and venue art pipeline works, and binary game assets (Content/) moved from git to Perforce on 2026-09-22. The project is on schedule for GDD Draft 2, due 19 October 2026 (section 20).

Report	Items
Complete	Preliminary project plan complete. Controller layout fixed on 2026-09-14 (section 7). Base fighter: MetaHuman body (175 cm, medium build) in seven wardrobe pieces: dobok jacket and trousers, belt, chest protector, helmet, gloves and foot guards, with Chung and Hong materials; passed a full-body range-of-motion test. Referee roster: six MetaHuman referees in uniform, with Chaos cloth on ties and trousers. Staging levels L_CompetitionArena (8 m octagon mat in a 12 m square) and L_TrainingDojang. Motion capture shots planned. Audio recordings planned: every voice line, sound effect, ambience and music cue, by milestone. GDD Draft 1 (this document) and its presentation slides (Docs/GDD_D1P_OpenStance.pptx): presented 28 September 2026, with instructor and second-reader feedback due 2 November (section 20). Binary game assets (Content/) migrated from git to Perforce on 2026-09-22, with exclusive-checkout locking for unmergeable UE5 binaries. MotionBuilder test character prepared for the BCIT Vicon Nexus motion-capture session (Resources/Models/MotionBuilder/OpenStance_TestFighter.fbx; Docs/Vicon_Nexus_Test.md); the preliminary lab appointment is complete.
In progress	GDD Draft 2, due 19 October 2026 (section 20). Skill Stick and motion capture proof of concept for Milestone 1, due 9 November (sections 19-20). BCIT Vicon Nexus motion-capture test session: the preliminary lab appointment is done; the capture session itself has not yet taken place (Docs/Vicon_Nexus_Test.md).
Blockers	None.
Debt and major issues	Active Defense, the roundhouse/hook stick direction and the Tornado Kick's modifier conflict with Fake Attack are now defined (7.2, 8.8, 18.2); they still need playtesting once Milestone 1 code exists. The runtime module Source/OpenStance has no gameplay code yet, so the Skill Stick stays untested until Milestone 1. Stamina, movement and hit-reaction values are starting values that have not been playtested. Second Half dates are not published.

1.3 Concept Statement

Real taekwondo sparring, one on one: read the distance, choose your kick with a flick of the Skill Stick, and land it before your opponent does.

A match lasts 5 to 8 minutes and one exchange lasts 1 to 3 seconds. One player spars the AI, or two players spar on one console.

1.4 Genre

3D one-on-one sports fighting game. From fighting games, Open Stance takes committed attacks with startup, active and recovery phases, reads, fakes and counters. From sports simulation, it takes a real ruleset: World Taekwondo scoring, electronic protectors, penalties and ranks. There is no health bar and no combo string; matches are won on points, as in the real sport.

1.5 Target Audience

Audience	Who	Games they play	What they want
Fighting-game players	Ages 16–35, controller players	Street Fighter 6, Tekken 8	Depth through spacing and timing rather than long combo strings; a high skill ceiling; local versus
Combat-sports game players	Ages 16–40	EA Sports UFC 5, Undisputed	A realistic sport with simulation controls and a career
Taekwondo practitioners	Athletes, coaches and club members, age 13 and up	Taekwondo Grand Prix, or none	Their sport shown accurately: real rules, techniques and etiquette
Martial arts schools **[Future]**	Clubs and coaches	—	Tactics teaching and remote sparring leagues

Motivations: mastery (a fair, readable system where skill decides the match), competition (winning on points against a friend or rival) and authenticity (real technique and rules).

ESRB target: E10+ (Everyone 10+), Mild Violence. Contact is full but sporting: no blood, no injuries shown and no taunts. The ESRB assigns the final rating; online interactions are not rated.

1.6 Concept and Unique Selling Points

A match is a series of short exchanges on an 8 m octagonal mat. Each fighter bounces in guard and manages distance with the left stick. The right stick, the Skill Stick, chooses the kick: up for a side or cut kick, toward the lead leg for a roundhouse kick or the other side for a hook kick, and circles and other motions for axe, spinning and tornado kicks (7.2). Holding RT kicks with the back leg, holding RB aims at the head, holding LB fakes whatever kick the gesture would throw, and holding LT reads the Skill Stick as Active Defense instead of offense (8.8). Hits score the way the electronic scoring system does: 1 point for a punch, 2 for a body kick, 3 for the head, 4 for a spinning body kick and 6 for a spinning head kick (8.3). Stamina limits how often a fighter throws big techniques. Players learn in the training dojang and spar the AI or a friend; Career

then takes a created fighter from white belt at a local club to a world final, unlocking techniques rank by rank. The look is realistic: MetaHuman fighters in white doboks and blue or red gear, animated from motion capture of real practitioners.

Unique selling points

- **The Skill Stick.** Kicks are right-stick gestures, and two held modifiers choose the leg (RT) and the target (RB), so a small set of gestures covers the sport's main kicks at body and head height.
- **Real sparring, real scoring.** World Taekwondo rules: 2-minute rounds, electronic protector scoring, 1 to 6 point techniques, gam-jeom penalties, and a 12-point gap.
- **Distance and commitment decide matches.** Attacks never home in from out of range, and every technique can be punished on recovery (1.7).
- **Motion captured from real practitioners.** Techniques, stance switches, reactions and falls come from taekwondo athletes.
- **Respect for the sport.** Real ranks, competition colours and etiquette, and nothing that affects a match is ever sold (1.7).

Prototype: none yet. The Milestone 1 proof of concept and its video link here when complete (section 19).

1.7 Design Pillars

Pillar	Meaning
Taekwondo first	Techniques and movement that work in real sparring. Nothing is added because it looks flashy in other fighting games.
Distance matters	Attacks don't magnetically connect from unrealistic range. Small spatial correction is fine; bad distance still misses.
Commitment matters	Startup, active and recovery phases create counter openings. Big techniques are big risks.
Score and stamina are separate	Score decides who is winning. Stamina decides what the fighter can execute well right now.
Animation serves gameplay	Mocap is cut into reusable states, transitions and reactions that gameplay drives.
Respect the sport	Real ranks, real competition colours, real skin, hair and eye colours. Skill decides matches, never purchases.

1.8 Player Experience

Question	Answer
Who is the player	A taekwondo athlete at a competition, in dobok and full sparring gear. In Career the same created fighter grows from a white-belt beginner to a black-belt international competitor.
Setting	The present-day sport: a training dojang and competition venues. There is no scripted plot and no fantasy setting (6.1).

Question	Answer
Fantasy	Being a sharp, composed sparring athlete: reading an opponent, landing the head kick at the right moment and winning under scoreboard pressure.
Emotions	Tension while circling at kicking distance; the snap of a clean score; pressure when behind on the clock; pride in a disciplined win; respect at the bow.
Length	A session: 20 to 40 minutes of lessons and matches. A Career season: 3 to 5 hours, with an estimated 30 hours to reach 3rd dan (level 40, 5.2). Match and round lengths: 2.4.
What keeps them playing	Every exchange is a new read, the score and clock change the right decision, and improvement shows as more points from fewer, better techniques.
Major phases	1. Learn: Training lessons for movement, the Skill Stick and scoring (first 15 minutes). 2. Compete: Sparring against the AI and local friends. 3. Master: harder AI, spinning and jumping kicks, clean matches without penalties. Career (v1.0) and Online **[Future]** extend Compete and Master (sections 4 and 12).

1.9 Key Moments

Moment	When	What happens
First score	Minutes 1–5	In a Training lesson the player pushes the right stick  sideways. The lead leg throws a roundhouse kick into the protector, the scoreboard shows +2 and the protector sound confirms it.
First read	First AI match	The AI advances; the player dashes back  and meets its approach with a side kick  . Timing and distance, not button speed, won the point.
The wall	A harder AI or a first match against a friend	The opponent cut-kicks every approach and the player falls 4 points behind. Fake attacks  and switching sides  get the player past the cut kick.
The comeback	Final round, last 10 seconds	Down by 3, the player fakes an attack, then throws a spin hook kick  +  for 6 points as the clock runs out.
The podium	Career: world series final	In a black dan dobok, the fighter bows to the rival who beat them two seasons earlier, then takes gold.

1.10 Art, Sound and Music

Visual style. Realistic and clean, like a broadcast of a World Taekwondo event. Fighters are MetaHuman bodies in white doboks with blue (Chung) or red (Hong) protectors and helmets. Arenas use bright, even light so limbs and scoring zones stay readable. The Chung-blue and Hong-red convention runs through the mat, gear, scoreboard and HUD, so the player always

knows which side is which. The training dojang is warmer: daylight, wood, and walls of kicking pads.

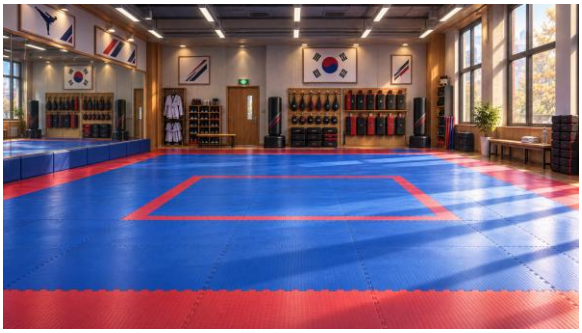
Reference images (Resources/References) set tone and layout. They are not used in the game.



Fighter: dobok, Chung protector and helmet, gloves and foot guards (front, side, back)



Competition arena: octagon mat, scoreboards, officials' tables, coach chairs



Training dojang: kicking pads, heavy bags, mirrors



Referee uniform

Sound. Sound carries information: the protector impact changes with the points scored (1 to 6), feet squeak and slap on the mat, and the referee calls commands in Korean with English subtitles, each of the six referees in their own voice (11.2). Fighters kihap (shout) as they kick, grunt when hit and breathe harder as their stamina drops below 25 (8.9). Both fighters in a match come from the same roster (7.5), so at least two men's and two women's voice sets keep them from sounding alike. Coaches shout short English cues from the corner chairs during play; advice between rounds stays as on-screen text. Crowd size and reverb grow with the venue, from a quiet club night to a full arena; crowds cheer, gasp and applaud, but never boo. Every voice stays within the E10+ rating (1.5): efforts and shouts, no pain screams and nothing that mocks a fighter.

Music. Music plays in menus, loading screens and round breaks only; rounds have no music, so impacts and footwork stay readable, and only a quiet pulse marks a round's last 10 seconds. Style: modern electronic tracks with Korean percussion. Whether music is commissioned or licensed is unresolved (18.1).

Asset lists. Every voice line, sound effect, ambience and music cue is planned and phased by milestone; a complete art asset list does not exist yet (18.1). Presentation details: section 13.

1.11 Current Target Platform

Item	Target
Platform	PlayStation 5 and Xbox Series X|S
Development platform	Windows 10 and 11, 64-bit PC. The First Half milestones (M1, M2; ending December 2026) are developed and demonstrated here. A console release needs a licensed developer account and devkits that the project does not have yet (18.1); PlayStation 5 and Xbox Series X|S remain the v1.0 target for the finished capstone in Spring 2027.
Input	Controller only: one per player, two for local versus. There is no keyboard or mouse play.
Engine and graphics	Unreal Engine 5.8; DirectX 12 on Xbox and on the Windows development build, the PlayStation 5 renderer on PS5
Performance	60 fps during matches: 1920 × 1080 on the Windows development build, 3840 × 2160 on PlayStation 5 and Xbox Series X, 1920 × 1080 on Xbox Series S
Performance verification	Resolution and frame rate measured on the Milestone 3 build (18.1)
Trophies and achievements	PlayStation trophies with a Platinum and Xbox achievements worth 1,000 Gamerscore, from one shared list (5.7)

A PC release on Steam and cross-play are in section 17.

1.12 Competition

Game	Release	Sales and budget	Description
Taekwondo Grand Prix Hello There Games	PC, 9 October 2018	Not published	The closest direct competitor: taekwondo sparring with World Taekwondo rules and scoring, motion-captured movement, local and online play and customization.
Virtual Taekwondo Refract Technologies with World Taekwondo	VR: PICO 4 and PICO 4 Ultra	Not published	World Taekwondo-endorsed VR sparring with body-worn trackers; the player's body is the controller. Used for the World Taekwondo Virtual Championships (Singapore, November 2024).
Tekken 8 Bandai Namco	PC, PS5, Xbox Series X|S; 26 January 2024	About 3 million copies by February 2025	3D fighting game with a taekwondo-based character (Hwoarang); health bars, juggles and combos.
Street Fighter 6 Capcom	PC, PS4, PS5, Xbox Series X|S; 2 June 2023	More than 7 million units by June 2026	2D fighting game with a taekwondo-based character (Juri) and simplified Modern controls that widen its audience.
EA Sports UFC 5 EA Vancouver	PS5, Xbox Series X|S; 27 October 2023	Not published	Mixed martial arts simulation with stand-up striking, stamina, a career mode and online play.
Undisputed Steel City Interactive	PC early access January 2023; PC, PS5, Xbox Series X|S 11 October 2024	1 million copies; about US$18.8 million raised for development	Licensed boxing simulation built on footwork, stamina and realistic punch selection.
Fight Night Champion EA Canada	PS3, Xbox 360; March 2011	Not published	Boxing game whose analog-stick punching is the closest precedent for choosing strikes with a stick gesture.

What sets Open Stance apart. Taekwondo Grand Prix, the only direct competitor, is PC only, and Virtual Taekwondo needs a VR headset and body trackers. No taekwondo sparring game ships on PlayStation or Xbox; Open Stance targets both. Open Stance differs from both through the Skill Stick, which gives controller players a readable, expressive way to choose kicks. It differs from Tekken 8 and Street Fighter 6 by using the sport's real rules and scoring instead of health bars, and from UFC 5 and Undisputed by bringing their simulation approach to a kicking sport neither covers.

Sources: Steam store and Metacritic (Taekwondo Grand Prix); virtualtkd.gg; Bandai Namco via Game Developer (Tekken 8); Capcom investor relations and Gematsu (Street Fighter 6); EA (UFC 5); Forbes and PlayStation Universe (Undisputed). Checked 2026-09-14.

1.13 Monetization

Phase	Model
v1.0	None. v1.0 is a free portfolio and demonstration build.
Future	A commercial release would be premium: a one-time purchase on the PlayStation Store and the Microsoft Store, with optional cosmetic content only (gear styles, dojang items, ceremonies). Never ranks, techniques, attributes or anything that changes a match. The price is unresolved (18.1).

A premium price suits a competitive sport game built on fairness (1.7): every player has the same techniques, and Ranked would give everyone the same attributes. Free-to-play progression would put purchases between the player and techniques, which the balance rules forbid (5.6).

1.14 Scope and Team

Item	Plan
Team	One developer (Joey Driedger): design, programming, technical art and production. Volunteer taekwondo practitioners perform the motion capture.
Length	About 8 months, in two halves: the First Half (September to December 2026) builds the sparring core and ends in playable demo gameplay, and the Second Half (January to spring 2027) adds Career and finishes the build.
First playable	Milestone 2, due 7 December 2026: two local players complete a basic sparring match. This is the demo gameplay that closes the First Half.
v1.0 content	premade men's and women's fighters (at least one of each), the Career fighter creator's five body types (5.4.2), the competition arena and training dojang, 8 to 12 captured techniques, Sparring, Training and local Tournament modes, an AI opponent, and Career seasons with training, grading, tournaments, rank-gated technique unlocks and its own venues. Online play, online tournaments, challenges, replays and video review are Future work (section 17).
Time to complete	Sparring and Training have no end state. Career runs about 30 hours to 3rd dan, the level 40 cap (5.2), and the world series final (4.2).
Major risks	Capture of fast kicks, animation workload, responsiveness of mocap-driven attacks, Skill Stick complexity and scope growth. Mitigations: 16.4.

2 Player Objectives and Progression

2.1 The Player

The player controls one taekwondo fighter in a one-on-one match, seen from a side-on camera. In Sparring the player picks a fighter; in Career the player creates one.

What the player knows at the start. Nothing about taekwondo is assumed. The first launch opens Training lesson 1, which explains stance, guard, distance and the scoreboard in under 3 minutes and can be skipped. There is no scripted plot; the season calendar, the rivals and the coach's notes carry the story (4.3).

2.2 Goals

Horizon	Goal	Shown by
Exchange: 1–3 s	Score a technique, or make the opponent miss and counter	Score pop-up and sound
Round: 2:00	Lead on points at time, or reach a 12-point gap	Scoreboard and clock
Match: 5–8 min	Win 2 rounds	Round indicators and results screen
Session	Finish lessons, raise drill grades, beat a harder AI	Lesson list, grades, difficulty list
Season	Pass grading exams and qualify for higher event tiers	Career calendar and ranking points
Career	Win the world series final; enter the hall of fame **[Future]**	Career record

Progression is linear only where rank gates it: Career event tiers and technique unlocks follow rank (5.3, 5.5). Everything else is the player's choice: which events to enter, what to train and how to fight.

2.3 Moment-by-Moment Gameplay

READ DISTANCE → SET UP → COMMIT → SCORE OR MISS → RESET

- **Read distance (0.5–2 s).** Both fighters bounce in guard just outside kicking range. The player steps and angles with the left stick  and switches sides with  , watching the opponent's lead leg and weight.
- **Set up.** A fake attack (hold ), a guard change on the D-pad  or a dash  draws a reaction.

- **Commit.** A Skill Stick gesture throws the technique, with RT for the back leg and RB for the head. Its startup, active and recovery phases (MoveData, 16.3) decide whether it lands before the opponent's counter.
- **Score or miss.** A valid hit on the protector or helmet scores on the HUD with a sound matched to the points. A miss leaves the attacker in recovery, open to a counter.
- **Reset.** Fighters return to guard; the referee calls kal-yeo to separate a clinch. The next read begins.

Controller layout: 7.1. Techniques and inputs: section 9.

2.4 Core and Outer Loops

Loop	Length	Cycle	Why it engages
Exchange (core)	1–3 s	See 2.3	Constant small decisions with immediate, readable feedback
Round	2:00	Exchanges → score changes → adjust risk to the score and clock	The clock and point gap change which risk is right
Match	5–8 min	Round → 1:00 break → round	Adapting to an opponent's habits between rounds
Session	20–40 min	Train a technique → spar → review results → train	Visible improvement in grades and results
Career week	10–20 min	See 4.1	Rank, rivals and events give long-term goals

Expected emergent play: players develop styles (counter fighter, pressure fighter, head-kick specialist) from one shared move list, and in Career a fighting style (5.4.3) supports them; they condition an opponent with repeated fake attacks before throwing the real kick; and they take late-round risks when behind. The loops contain no monetization points (1.13).

2.5 Interactivity

Type	In Open Stance
Action and feedback	Stick gestures and held modifiers map directly to techniques, legs and targets. The score appears the moment a hit registers, with a protector sound and reaction matched to the points.
Short-term cognitive	Reading distance, stance (open or closed) and the opponent's guard; choosing body or head, fake or commit, in under a second.
Long-term cognitive	Learning an opponent's habits across rounds; managing stamina and the point gap over a match; planning training and events in Career.
Emotional	Tension at kicking range, relief at a block, pride in a clean spinning kick, respect at the bow.
Social	Local versus on one console; online casual and ranked **[Future]**; clubs and team events **[Future]**.

Type	In Open Stance
Cultural	The sport's etiquette, Korean commands and ranks; a way for practitioners to share their sport with people who have never trained.

2.6 Progression Through the Game

TRAINING LESSONS → SPARRING VS AI → LOCAL VERSUS → CAREER

The player starts with the six Training lessons (section 3), then spars the AI at rising difficulty and plays friends. Outside Career every technique is available from the start. The Second Half adds Career, with rank-gated progression and technique unlocks (sections 4 and 5); Online play is Future work (section 12). Key moments along this path: 1.9.

3 Game Modes

3.1 Play

Sparring. A single match with no progression attached.

- Player vs player (local) or player vs AI.
- Settings: ruleset options (8.2), arena, AI difficulty, fighters.
- All techniques unlocked; Equal Terms attributes by default (5.6).

Tournament. Single-elimination brackets of 8, 16 or 32, played locally. AI fighters fill every slot the players do not take, so one player can run a full bracket alone. Several local players can enter one bracket.

Team Championship **[Future]**. Club or national teams in a tag-team format modelled on WT team events: fighters rotate in and out during a bout, and a shared team score decides the match. Up to three local players per team.

Challenges **[Future]**. Short scenarios with goals, such as “Win the round from 5 points down with 30 seconds left” or “Score a spin hook kick right after switching sides”. Medals unlock cosmetics.

3.2 Progress

Training (도장 dojang)

- **Lessons:** Six guided tutorials: movement; Skill Stick kicks; back leg, head and fake modifiers; switching sides; guards and active defense; stamina.
- **Drills:** Kicking paddles, reaction targets and switch-side timing, with grades.
- **Free practice:** Dummy or AI partner, record and playback, input display, debug overlay.

Career. Take a created fighter from white belt at a local club to world-level events. See section 4.

3.3 Other Sparring Formats

Para taekwondo **[Future]**. Para gyeorugi under WT Para Taekwondo rules (for example, no kicks to the head), with animation captured with para athletes and classification-accurate fighters. Developed with para practitioners, not approximated.

3.4 Connected

Online **[Future]**

- **Casual:** Quick match and private rooms.

- **Ranked:** seasons with a skill rating; Equal Terms always on. See section 12.

Replays [Future]. Saved matches replayed from inputs and scoring events, with free camera, slow motion and clips of scoring exchanges. Photo mode and sharing [Future].

Broadcast [Future]. Spectator and commentator tools for community tournaments: director camera, live scoreboard overlays, instant replay and bracket display.

3.5 Referee Mode [Future]

The player takes the centre referee's place on the mat and officiates a match between two fighters: two AI fighters, two local players, or an online match [Future]. Scoring stays with the electronic protectors (8.3). The referee player runs the match: starting and stopping play, separating clinches, declaring penalties, counting knockdowns and raising the winner's hand, with the commands in 8.5.

- **Calls decide the match.** Only the penalties the referee player declares are applied. The automatic penalty detection (8.4) still runs, but only as the answer key for grading; it never gives a gam-jeom itself.
- **Referee grade.** Each match ends with a grade for call accuracy (correct, missed and wrong penalties), call timing, and positioning: keeping both fighters in view without blocking the camera (11.2). The rating meter (3.5.1) tracks the same calls live.
- **Camera.** An over-the-shoulder referee camera that follows the exchange; the side-on combat camera (section 13) is an option.
- **Players.** Up to three controllers in local play: Chung, Hong and the referee. The fighters keep the layout in 7.1; the referee has its own layout, below.
- **Learning.** An extra Training lesson, beyond the six in 3.2, teaches the commands and the penalty list before the first match.

Referee controls

Action	Input
Move on the mat	 Left stick
Turn and look	 Right stick
Select Chung (blue, left)	 LB
Select Hong (red, right)	 RB
SHI-JAK (start) or KYE-SOK (continue)	 A
KAL-YEO (break; stops play)	 B

Action	Input
GAM-JEOM to the selected fighter	 Hold X, choose the reason with RS, release
KEU-MAN (stop; end of round)	 Y
Knockdown count	 Hold LT: counts one to eight in Korean; release to stop
Raise the selected fighter's hand	 RT, after the final KEU-MAN
CHARYEOT and KYEONGRYE (attention, bow)	 D-pad Up
JOON-BI (ready)	 D-pad Down
Pause	Menu button

The gam-jeom reasons on the right-stick wheel follow 8.4: boundary, falling, avoiding or delaying, leg lifting, kicking below the waist or punching the head, and grabbing or pushing. Like the other Korean commands, the knockdown count is shown with English subtitles.

3.5.1 Referee Rating [Future]

A rating meter shows how well the referee player is officiating. It starts every match at 100 and moves with each call, judged against the automatic penalty detection (8.4). The values below are starting values, tuned in playtests like the stamina values in 8.9.

Event	Rating change
Correct call: right command, right fighter, on time	+2
Late call, such as KAL-YEO more than 2 s into a clinch	−3
Missed penalty	−6
Wrong penalty: wrong fighter, or no foul	−10
Missed knockdown count	−10
Blocking the camera, or losing sight of both fighters for more than 3 s	−2 each time
Call overturned by video review [Future] (8.11)	−8

Wrong calls cost more than missed ones: a wrong gam-jeom changes the score, a missed one does not.

- **Display.** A bar on the referee's HUD that turns amber below 40, like the stamina bar (7.6). In local play the fighters see it too.
- **Warning.** Below 40, the technical delegate, an off-screen official, sends a warning to the referee's HUD.

- **Removal.** Below 20, the technical delegate stops the match at the next break in play and an AI referee takes over for the rest of the match. The player's match grade is F. The score stands: calls already made are not corrected, as in the real sport.
- **Only the rating removes a referee.** Fighters cannot vote a referee out, so removal cannot be used to grief, and the sport is respected (1.7). Fighters challenge calls through video review **[Future]** (8.11), and an overturned call lowers the rating.
- **Online referees** **[Future]**. Referees for community tournaments (3.4) carry a public rating across matches; a low rating takes them off an organizer's referee list.

4 Career

Career follows one created fighter through seasons of training, grading and competition. A season calendar sets training slots, grading dates and the events the fighter enters; tournaments run in several formats (4.2), and ranking points decide which tier opens next. Techniques unlock by rank and are learned in a drill (5.5), so the move list grows with the fighter. There is no scripted plot; rivals, gradings passed and titles won are the record (6.1). Career menus show the fighter out of match gear: in dobok and belt, at the club, on the podium.

4.1 Weekly Loop

PLAN THE WEEK → TRAIN → PREPARE → GRADE OR COMPETE → REVIEW → LEVEL UP

- **Plan:** A season calendar of training slots, rest days, grading dates and the events the fighter has entered.
- **Train:** Drills and sparring earn XP and attributes, and a newly unlocked technique is learned in a drill before it can be used (5.5). Training without rest lowers readiness for the next event.
- **Prepare:** Before an entered event the coach scouts the bracket and opens drills aimed at one opponent's habits; the last week is a taper that trades XP for readiness.
- **Grade:** Belt grading exams gate rank (5.3).
- **Compete:** Events award ranking points, XP and unlocks.
- **Review:** Points by technique, stamina use and penalties, with a coach's note pointing to a drill.

4.2 Event Tiers

Event names are fictional, so no real competition is licensed. Tiers differ in format and venue (6.2) as well as size. A club night is a round robin. The opens and championships are single-elimination brackets, seeded on ranking points from tier 4 up. The world series final splits its eight qualifiers into two pools before the knockout.

Tier	Event	Entry	Format	Bracket
1	Club sparring night	Any rank	Round robin	4
2	City open	Yellow belt	Single elimination	8
3	Provincial championship	Green belt	Single elimination	16
4	National championship	Blue belt, provincial podium	Single elimination, seeded	16
5	Continental open	Black belt, national ranking	Single elimination, seeded	32

Tier	Event	Entry	Format	Bracket
6	World series final	Top 8 in ranking points	Two pools, then knockout	8
7 **[Future]**	Quadrennial world games	Qualification over a 4-season cycle	Single elimination, seeded	16

4.3 People

- **Coach:** Weekly suggestions, opponent scouting before events, short shouts from the corner chair during matches (1.10), post-match notes.
- **Rivals:** Recurring AI fighters with their own styles who rise through the tiers alongside you.
- **Club mates:** Sparring partners who unlock drills and friendly matches.

4.4 Long-Term Career [Future]

- **Age divisions:** Start as a cadet (12–14, poom rank, helmets with face shields) or junior (15–17) and move up to senior.
- **Ageing:** Peak years, gradual attribute decline and retirement over a long Career.
- **Fatigue and minor injuries:** Knocks that affect readiness, handled with rest and physio choices; never permanent or graphic.
- **Hall of fame:** Retired fighters' records, medals and signature techniques.

5 Progression and Levelling

Two tracks run side by side. Fighter level comes from XP and grows attributes. Rank (belt and dan) is earned at grading exams and gates events, techniques and dobok style. Neither track lets stats beat skill.

- **Match results screen with points by technique**
- **Fighter level, XP, attributes, ranks and technique unlocks (Career)**
- **PlayStation trophies and Xbox achievements (5.7)**
- **Ranked seasons and skill rating (Online) [Future]**
- **Ageing, retirement and hall of fame [Future]**

5.1 XP Sources

Source	XP	Notes
Match win	300	×1.0–2.0 by event tier
Match loss	150	Losing still teaches
Round won	50	Per round
Technique variety	≤100	10 per distinct scoring technique in a match
Clean match (no gam-jeom)	50	Rewards discipline
Training drill	40–120	By grade achieved
Grading exam passed	500	Once per rank

5.2 Level Curve

XP to go from level L to L + 1 is 250 + 75 × L. The level cap is 40, and each level grants 2 attribute points. Career drills add attribute points too (4.1). Points raise each attribute up to its cap (5.4.1).

Level	XP to next	Total XP to reach	Roughly
2	400	325	First session
5	625	1,750	First city open
10	1,000	5,625	Green belt range
20	1,750	19,000	Red belt range
30	2,500	39,875	1st dan range
40	—	68,250	Cap

5.3 Ranks

Belt colour order varies between schools; this is the game's standard. A grading exam is a set of technique trials plus a graded sparring bout, available once the level gate is met.

Rank	Belt	Collar	Level gate	Unlocks
10th–9th 급 gup	White	White	1	Tier 1 events
8th–7th gup	Yellow	White	4	Tier 2 events
6th–5th gup	Green	White	8	Tier 3 events
4th–3rd gup	Blue	White	14	Tier 4 events
2nd–1st gup	Red	White	20	Advanced drills
1st 단 dan	Black	Black	28	Tier 5 events, dan dobok
2nd dan	Black	Black	34	Signature technique slot
3rd dan	Black	Black	40	Tier 6 seeding bonus; second signature technique slot

Under-15 fighters hold 품 poom rank with a red-black belt and collar (cadet careers, 4.4).

5.4 Attributes

Attributes start from a floor of 40, with 70 as baseline. Most cap at 99, but a fighter's specialized attributes can pass 100 (5.4.1). Each point away from 70 changes an attribute's effect by one third of a percent: 40 is −10%, 100 is +10%, and the highest possible rating, 108, is about +13%. Effects stay small, so good timing from a low-stat fighter beats bad timing from a high-stat one.

Attribute	Affects
Power	Impact margin over the trunk registration threshold; reaction severity
Speed	Startup and recovery time of techniques
Flexibility	Head-kick height tolerance
Stamina	Size of the stamina pool
Recovery	Stamina regeneration, including between rounds
Footwork	Step distance and dash cost
Technique	Target zone tolerance and fake-attack cancel window
Composure	Stamina drain under pressure; passivity timer leniency

5.4.1 Overall and Potential

A fighter's overall rating (OVR) is the average of the eight attributes, rounded down. Under Equal Terms every fighter's OVR is 70. A Career fighter starts with every attribute at 64, then applies the modifiers for its body type (5.4.2) and fighting style (5.4.3). The modifiers in each

row add up to zero, so every Career fighter starts at OVR 64: body type and style change the shape of the ratings, not the total, and never the potential.

- **Potential.** Every Career fighter can reach OVR 99, whatever the body type and style. Each attribute caps at 99 plus that attribute's body type and style modifiers, so specialized attributes pass 100 and weak ones cap lower. The modifiers add up to zero, so the caps always average exactly 99: OVR 99 means every attribute at its cap, the mark of a complete Career.
- **Growth.** Level-up points (5.2) and Career drills (4.1) raise attributes up to their caps, tuned so that a fighter who trains every week reaches OVR 99 around the level 40 cap. Once every attribute is at its cap, no more points are awarded.
- **Fairness.** A Lean tall Head-kick specialist caps Flexibility at 108 and Technique at 101 but Power at 94 and Stamina at 92. Even at 108, an attribute's effect is only about 13% above baseline (above), so timing still decides exchanges.

5.4.2 Body Types

Each body type has a set height and weight, built on a MetaHuman body (10.1). Height sets reach (6.4). Each body type raises some attributes and lowers others by the same total, so no body type is stronger overall: taller bodies gain reach but give up other attributes (Tall loses Speed and Footwork, Lean tall loses Power and Stamina). The same modifiers move each attribute's cap (5.4.1).

Body type	Height, weight	Base rating modifiers
Compact	165 cm, 60 kg	Speed +4, Footwork +2, Recovery +1, Power −3, Flexibility −4
Medium	175 cm, 68 kg	None (the base fighter, 1.2)
Stocky	172 cm, 80 kg	Power +4, Stamina +3, Speed −2, Flexibility −5
Lean tall	185 cm, 72 kg	Flexibility +4, Footwork +1, Power −2, Stamina −3
Tall	192 cm, 84 kg	Flexibility +3, Power +3, Speed −3, Footwork −3

The body type is chosen in the fighter creator and cannot change during a Career.

5.4.3 Fighting Styles

A fighting style shifts base ratings toward one way of sparring. Like body types, styles move points between attributes without changing the total, so every style has the same starting OVR of 64 and the same OVR 99 ceiling; only the attribute caps move.

Style	Plays like	Base rating modifiers
All-rounder	No weak spot; adapts to each opponent	None
Counter fighter	Waits at range and punishes recovery	Composure +4, Technique +2, Footwork −3, Stamina −3

Style	Plays like	Base rating modifiers
Pressure fighter	Keeps moving forward and forces exchanges	Stamina +4, Footwork +3, Composure −3, Flexibility −4
Head-kick specialist	Hunts 3- and 6-point head kicks	Flexibility +5, Technique +2, Power −3, Stamina −4
Cut-kick blocker	Stops every approach with the lead-leg cut kick	Speed +3, Composure +2, Power −2, Flexibility −3

The style is chosen in the fighter creator. It can change once per season, at the start of the season with the coach (4.3); changing it moves the modifiers, not the points the fighter has earned.

- **Equal Terms.** Sets every attribute to 70 and ignores body type and style modifiers (5.6). Reach from the body's height still applies.
- **AI fighters.** Rivals and AI opponents use the same body types and styles; an AI fighter's tactics (11.1) follow its style.

5.5 Technique Unlocks

In Career, techniques unlock by rank and are then learned through a drill. The back-leg and head modifiers arrive with the first technique that needs them. Every other mode has the full move list from the start (section 9).

Rank	Techniques
White	Movement, guards, punch, side and cut kick, roundhouse kick to the body
Yellow	Active defense, fake attack, dash, switching sides
Green	Roundhouse kick to the head, spin side kick
Blue	Hook kick, axe kick
Red	Spin hook kick, jump kicks (side, roundhouse, hook, axe, spin hook)
Black	Tornado kick, signature variations

5.6 Balance Rules

- **Equal Terms: all attributes set to 70. The default in Sparring and Tournament; always on in Ranked [Future].**
- Appearance customization (face, skin, hair, eyes and gear) never changes attributes or point values. Body type and fighting style shift Career base ratings between attributes without changing OVR or the OVR 99 ceiling (5.4.1); Equal Terms overrides them.
- Level, rank and techniques are earned in play. They are never sold, in any phase.

5.7 Trophies and Achievements

Open Stance has one list of 35 goals, released as PlayStation 5 trophies and Xbox Series X|S achievements with the same names and conditions. On PlayStation each carries a grade (19 Bronze, 10 Silver, 6 Gold), and earning all of them unlocks the Open Stance Platinum. On Xbox they are worth 1,000 Gamerscore (G) in total, the standard for a full game. The list follows the path in 2.6: Training, then Sparring and Tournament, then Career.

Trophy / achievement	How to earn it	Nod to	PS5 trophy	Xbox G
Training				
Pure and Without Knowledge	Finish Training lesson 1.	ATA White Belt Meaning	Bronze	10
I Know Kung Fu	Finish all six Training lessons (3.2).	The Matrix (1999)	Bronze	20
No Words More Harmful Than Good Job	Earn the top grade in any Training drill.	Whiplash (2014)	Bronze	15
Wax On, Wax Off	Earn the top grade in every Training drill.	The Karate Kid (1984)	Silver	35
Sparring and Tournament				
Sweep The Leg	Win a match against the AI.	The Karate Kid (1984)	Bronze	15
Headshot	Score a 3-point kick to the head.	Shooter games	Bronze	15
Maximum Effort	Score a 6-point spinning kick to the head.	Deadpool (2016)	Bronze	20
Do a Barrel Roll	Score with a tornado kick.	Star Fox 64 (1997)	Bronze	20
Show Me Your Moves	Score with five different techniques in one match.	Super Smash Bros. (1999)	Bronze	20
This Is the Way	Win a match without receiving a gam-jeom.	The Mandalorian (2019)	Bronze	15
Twelve Points to Your House	Win a round by the 12-point gap.	Harry Potter House Points	Bronze	20
Eye of the Tiger	Win a match after trailing by 5 or more points in the final round.	Rocky III (1982)	Silver	30
Moment 37	Score the match-winning points in the last 5 seconds of the final round.	Evo Moment 37 (2004)	Silver	30
Git Gud	Win a match against the AI at its hardest difficulty.	Dark Souls players	Silver	40
Flawless Victory	Win a match against the AI at its hardest difficulty without conceding a point.	Mortal Kombat (1992)	Gold	50
Tournament Arc	Win a local Tournament of 8.	Anime tournament arcs	Bronze	20

Trophy / achievement	How to earn it	Nod to	PS5 trophy	Xbox G
There Can Be Only One	Win a local Tournament of 32.	Highlander (1986)	Silver	40
Here Comes a New Challenger	Finish a local versus match between two players.	Street Fighter II (1991)	Bronze	10
Career				
Ready Player One	Create a fighter in the fighter creator.	Ready Player One (2011)	Bronze	10
The First Rule of Fight Club	Win a club sparring night (tier 1).	Fight Club (1999)	Bronze	15
There's Always More To Learn	Pass a grading exam.	ATA Grandmaster Soon Ho Lee	Bronze	15
King of the Streets	Win a city open (tier 2).	Marco Ruas, UFC 7 champion (1995)	Bronze	20
The Champ Is Here	Win a provincial championship (tier 3).	Muhammad Ali	Silver	30
My Ring, My Rules	Win a national championship (tier 4).	Play on "My house, my rules"	Silver	40
Planting Seeds for the Future	Reach 1st dan.	ATA black belt meaning	Silver	40
The Continent's Most Wanted	Win a continental open (tier 5).	Madagascar 3 (2012)	Gold	50
We Are the Champions	Win the world series final (tier 6).	Queen (1977)	Gold	80
Final Form	Reach 3rd dan, at the level 40 cap.	Dragon Ball Z	Gold	60
Perfectly Balanced	Raise a Career fighter to OVR 99 (5.4.1).	Avengers: Infinity War (2018)	Gold	70
Gotta Learn 'Em All	Learn every Career technique in its drill, up to the tornado kick (5.5).	Pokémon	Silver	30
Do or Do Not	Finish a drill that a coach's note recommends.	The Empire Strikes Back (1980)	Bronze	10
Training Montage	Enter an event at full readiness after a taper week (4.1).	Sports films of the 1980s	Bronze	15
Now We're Even	Hidden. Beat a rival who beat you at an earlier event.	Rocky II (1979)	Silver	30
Get Me In the Hall Of Fame	Win every event entered in one Career season, with at least three events.	The Script, "Hall of Fame" (2012)	Gold	50
Easter eggs				
Up, Up, Down, Down	Hidden. Enter the Konami Code on the title screen (6.5).	Konami Code	Bronze	10
All trophies				
Open Stance	Earn every other trophy. PlayStation only; Xbox has no equivalent.	—	Platinum	—

Design rules. The list rewards what the game rewards: skill, discipline and a Career well run.

- **Earned in play, never bought.** No trophy needs purchased content, and a trophy gives nothing in game; cosmetic rewards come from Career (10.4, 1.13).
- **All in v1.0, all offline.** Every trophy can be earned in v1.0 without a network connection, so the Platinum and the 1,000 G are reachable in the capstone build and never depend on servers staying up. Future modes (Online, Challenges, Referee Mode and Team Championship) add their own trophies when they ship, as a separate PlayStation trophy group and an Xbox title update, never in the base list.
- **No farming with a second controller.** Match trophies count against the AI and in Career. A match between two local players, including one inside a Tournament bracket, counts only for Here Comes a New Challenger.
- **Nothing missable.** Every Career trophy can be earned in any season and any Career slot, and no body type or fighting style rules one out: every Career fighter can reach OVR 99 (5.4.1).
- **Respect the sport.** No trophy rewards penalties, losing on purpose or unsporting play (1.7). Two trophies are hidden: Now We're Even, because it names a Career rival, and Up, Up, Down, Down, because it gives away an easter egg.
- **Pop culture names, sport milestones kept plain.** Most names nod to films, games, combat sports and fighting-game culture, shown in the Nod to column. Three rank trophies take their names from the ATA, two of them from its Songahm belt meanings, in which the white belt is a seed and the black belt plants seeds for the future. A name is a short title or catchphrase only: no logos, artwork, music, song lyrics or characters, and nothing violent or mocking, to fit the E10+ target (1.5). Every reference needs a legal check before release (18.1).
- **Equal Terms either way.** Sparring and Tournament trophies count with Equal Terms on or off (5.6).

How they are built. An achievement subsystem listens to match, Training and Career events and checks each condition; trophies that build up over time, such as Wax On, Wax Off, keep their progress in the save (16.6). A DT_Achievements data table holds one row per trophy: its ID, String Table keys for the name and description (section 14), PlayStation grade, Gamerscore, hidden flag and icon. Unlocks go to PlayStation Network and Xbox network through Unreal Engine's online achievements interface. The Windows development build records unlocks in the save and shows a debug notice instead, so every condition can be tested before devkits exist (1.11, 18.1). Development builds with test cheats on (16.8) never report an unlock. Each trophy needs its own icon, listed with the art assets (1.10).

6 Game World

6.1 World Fiction

Open Stance is set in the present-day world of competitive taekwondo sparring. There is no invented backstory: the fiction is the sport itself, with its rules, ranks, etiquette and events. Clubs, event names and rival fighters are fictional so no real organization is licensed (4.2). The player meets this world through the venues and the referee and, in Career, through the coach, rivals and calendar. No separate world-fiction document is needed. Career's writing is calendars, rankings, rival records and coach's notes, not a script.

6.2 Venues

Venue	Unreal level	Used for	Look and crowd
Training dojang	L_TrainingDojang	Training lessons, drills and free practice. The first area a new player sees.	Warm daylight, wood floor, walls of kicking pads and mirrors. No crowd.
Competition arena	L_CompetitionArena	The default venue for Sparring and local Tournament matches, and for any Career event whose own venue is not built yet.	Neutral broadcast set: four scoreboards, officials' tables, bright even light. Small crowd.
Club dojang	L_ClubDojang (planned)	Career tier 1, the club sparring night (4.2).	A club's own training hall. Club mates on benches along the wall, one portable scoreboard.
School gym	L_SchoolGym (planned)	Career tier 2, the city open.	Basketball-court gym with folding bleachers and a few hundred spectators.
Provincial hall	L_ProvincialHall (planned)	Career tier 3, the provincial championship.	Sports hall with several mats running at once and a busy warm-up area.
National arena	L_NationalArena (planned)	Career tier 4, the national championship.	One raised mat in an arena bowl with video screens and a full lower tier of seats.
Continental arena	L_ContinentalArena (planned)	Career tier 5, the continental open.	Larger arena with the competing nations' flags, broadcast cameras and a louder crowd.
World-final stage	L_WorldFinalStage (planned)	Career tier 6, the world series final.	Darkened arena with a spotlit raised mat, a full crowd and an entrance walk.
World games stadium **[Future]**	L_WorldGamesStadium (planned)	Career tier 7 **[Future]**, the quadrennial world games.	The biggest venue: opening ceremony, podium and the largest crowd.

Players move between venues through menus; there is no traversal between locations. Every Career tournament tier has its own venue, and Career sets it from the event tier; crowd size, lighting and presentation grow with the tier (1.10). The venues share the contest area and props built by Tools/Unreal/build_arenas.py and differ in set dressing, lighting, crowd and scoreboards, so each new venue is set dressing, not new systems. Sparring and Tournament setup can choose any built venue (7.5).

6.3 Contest Area

Element	Specification
Mat	Blue octagon 8 m across, inside a red safety area; 12 m × 12 m in total (built by Tools/Unreal/build_arenas.py)
Boundary	The octagon's edge. A fighter with both feet over it receives a gam-jeom (8.4).
Start marks	Chung and Hong marks on the centre line, facing each other. Spacing follows the current WT rules (18.1).
Officials	Referee on the mat; judges' and recorder tables and four scoreboards around it, two on each side; a coach chair in the blue and red corners (1.10). Smaller Career venues scale this down: the club dojang has one portable scoreboard (6.2)
Camera	Side-on combat camera (section 13)

6.4 Movement and Physics

Movement comes from captured footwork, so step lengths and speeds follow real athletes. The values below are starting values for the Milestone 2 prototype; measurements from the Milestone 1 capture replace them.

Movement	Starting value
Bounce in place	2 bounces per second
Step (tap)	40 cm
Advance or retreat (hold)	1.2 m/s
Dash B	1.0 m in 0.3 s; closing to contact range starts a clinch automatically (above)
Side step or circle	35 cm per step
Back-leg roundhouse kick reach to the body	0.9–1.2 m between fighters' centres for the 175 cm base fighter; scales with body height (5.4.2)
Jump height Y	40–60 cm
Gravity	Unreal default, 980 cm/s²
Fighter collision	Capsule radius 35 cm; fighters never overlap, and chest-to-chest contact starts a clinch — whether both fighters press forward into it or one closes the gap with a Dash (9.3) that lands them in contact range

6.5 Easter Eggs

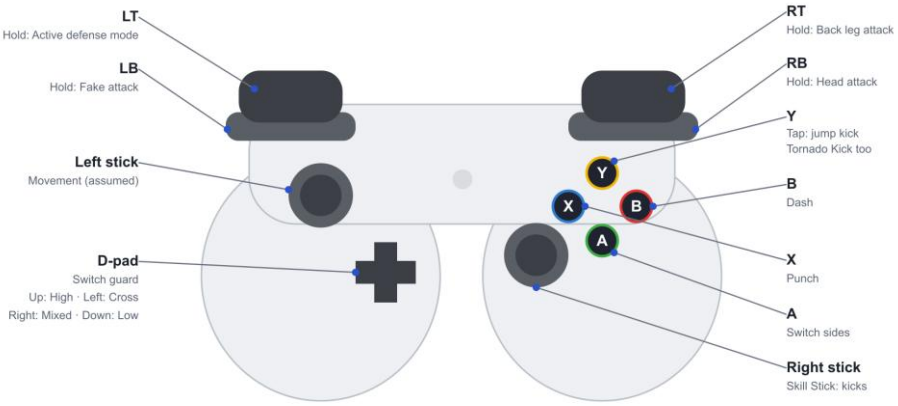
Six hidden details reward players who look closely. Easter eggs are cosmetic only: none changes a match, score or attribute (5.6), and each fits the E10+ target (1.5). Pop culture references follow the naming rules for trophies (5.7).

Easter egg	Where	Trigger	What happens	Nod to
Konami Code	Title screen	Up, Up, Down, Down, Left, Right, Left, Right on the D-pad, then B, A	The scoreboard and HUD switch to a pixel-art arcade style until the game closes, and the Up, Up, Down, Down trophy unlocks (5.7).	Konami games, from Gradius (NES, 1986)
Wax On, Wax Off	Training dojang, free practice	Leave the AI partner idle for 60 seconds	The partner picks up a rag and wipes the mirrors in slow circles, one hand then the other. Any input ends it. Needs one extra motion capture shot.	The Karate Kid (1984)
Bullet Time	Any venue	Win a match with a 6-point spinning head kick in its final second	Instead of the usual 1.5 s cut (section 13), the camera orbits the kick 360° in slow motion.	The Matrix (1999)
Crowd Signs	National arena and larger venues (6.2)	Random, after certain scores	A fan in the crowd raises a sign: DO A BARREL ROLL! after a tornado kick, IT'S OVER 9000! after a win by the 12-point gap. Signs cheer; they never mock a fighter.	Star Fox 64 (1997); Dragon Ball Z
Not Helpful Tips	Loading screens	1 loading screen in 50	A joke tip in place of a real one, for example "Kicks to the head score 3 points. Kicks from the head do not."	—
Capstone Club	Club dojang (6.2)	Match intro at a club sparring night	The intro camera pans past a banner for the BCIT capstone club and a team photo of the volunteer practitioners who performed the motion capture (1.14), as thanks.	This project

7 User Interface

The controller layout is below. In menus, A confirms and B goes back.

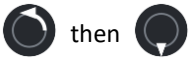
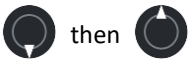
7.1 Controller Layout



Inputs are written in Xbox button names; the PlayStation equivalents map one to one. The left stick moves the fighter (7.4).

Input	Function
Left stick	Movement (assumed)
Right stick	Skill Stick: kicks (7.2)
LT Hold	Active Defense Mode
RT Hold	Back Leg Attack
LB Hold	Fake Attack
RB Hold	Head Attack
Y	Jump kicks: tap, then a kick's motion, including the Tornado Kick's full circle (9.1, 9.2)
X	Punch
A	Switch Sides
B	Dash
D-pad	Switch Guard (7.3)

7.2 Skill Stick Moves

Move	Input	Notation
Body Side / Cut Kick		RS Up
Roundhouse and Hook Kick		RS Left/Right: toward the lead leg for roundhouse, the other side for hook — mirrored when RT selects the back leg (18.2)
Axe Kick		RS Left/Right Quarter Circle Forward, Down
Spin Side Kick		RS Down, Up
Spin Hook Kick		RT + RS Left/Right Half Circle Down
Tornado Kick		Y, then RS Left/Right Half Circle Down, then Half Circle Up

Arc icons show one side; each circle motion works from the left or the right.

7.3 Guards

Guard	Input
High Guard	 D-pad Up
Cross Guard	 D-pad Left
Mixed Guard	 D-pad Right
Low Guard	 D-pad Down

7.4 Input Notes

- **Modifiers combine with the Skill Stick.** Hold  for the back leg and hold  for the head, for example hold  +  for a head roundhouse kick. Kicks without  use the lead leg.
- **The left stick decides standing versus sliding.** Leaving the left stick neutral while a kick or punch is thrown keeps the fighter rooted in place, the standing version of the technique. Holding the left stick in a direction through the technique instead slides the fighter that way during it — forward to close distance on the strike, back to step out

while still attacking — at the technique's normal step speed (6.4); the technique itself, its target and its points don't change.

- **Legs follow stance.** After Switch Sides  , the lead and back legs swap and the same inputs still work; the game resolves the physical leg at runtime.
- **Roundhouse and hook resolve by the leg that throws them.** RS toward the side of the leg throwing the kick is the roundhouse; the other side is the hook (7.2, 9.1). In Right Lead that's RS Right for a lead-leg roundhouse and RS Left for a lead-leg hook, mirrored in Left Lead; holding RT for the back leg mirrors both again, since the back leg is on the opposite side (8.7).
- **Fake and jump are modifiers, not separate moves.** Holding LB through any kick's gesture fakes that specific technique; holding LB alone with no gesture is a generic fake attack (2.3, 9.3). Y is a jump tap, not a held modifier. The right thumb can't hold Y and steer the Skill Stick at once, so tapping Y just before a side, roundhouse, hook, axe or spin hook kick's gesture throws it airborne (9.1, 9.2); without the tap, the same gesture throws grounded. The Tornado Kick uses the very same tap as every other jump kick; a full circle starting downward is what makes it the hardest input in the game, on purpose (9.2).
- **Distance still decides.** The input picks the technique; it never pulls the fighter into range (1.7).
- **Accessibility:** Hold-to-toggle for the four held modifiers **[Future]** (7.9).

7.5 Screen Flow

TITLE → MAIN MENU → MODE SETUP → FIGHTER SELECT → MATCH → RESULTS

Screen	Contents and flow
Title	Press any button to start. The screen shows only the game name, Open Stance: Gyeorugi, and that prompt. The first launch goes to Training lesson 1.
Main menu	A hub with tabs, switched with LB and RB: Play (Sparring, Tournament; Challenges and Referee Mode [Future]), Progress (Career, Training), Connected (Online and Replays) [Future], Options (Options, Controls, Quit)
Sparring setup	Players (1P vs AI or 1P vs 2P), ruleset (8.2), AI difficulty, arena
Tournament setup	Bracket size (8, 16 or 32), ruleset (8.2), AI difficulty, arena. The bracket view shows results and the next pairing between matches.
Fighter select	Sparring and Tournament use premade fighters only. The player picks the Men or Women roster, then a fighter for each side; either roster can supply the opponent. The side (Chung or Hong) is assigned, never chosen. The created fighter is used in Career only.
Match	Intro ceremony, rounds with the HUD (7.6), 1:00 breaks
Pause	Resume, move list, controls, options, restart, quit match; move list and controls open the controls reference (7.10) and return to the pause menu

Screen	Contents and flow
Results	Winner, round scores, points by technique, penalties; rematch, change fighters or main menu
Training	Lessons, drills with grades, free practice (3.2)
Career	Season calendar, training, event entry, grading, post-match review and the fighter creator (sections 4 and 10)
Fighter creator	Reached from Career only. Sex, body, fighting style, appearance and a gear on and off preview (section 10)
Options	Settings in 7.8
Controls	The controls reference in tabs (7.10), opened from Options, the Options Controls tab or the pause menu

A clickable prototype and a Lucid wireframe sheet of every screen are in the project (18.1).

7.6 HUD

Element	Position	Contents
Scoreboard	Top centre	Chung score in blue on the left, Hong in red on the right, round and clock in the centre, modelled on the PSS display
Penalties	Under each score	Gam-jeom count for that side
Stamina	Under each fighter's name	One bar per fighter; turns amber below 25
Score pop-up	Above the scoring fighter	Points scored (+1 to +6), shown for 1.0 s
Referee subtitle	Lower centre	Korean command and English meaning, for example KAL-YEO: Break
Input display and debug overlay	Left and right edges	Training only: controller icons for the last input, leg, move, target, stamina, hit result
Video review cards **[Future]**	Beside each score	One card per corner

7.7 Help System

- **Training lessons:** Step-by-step tutorials whose prompts show the controller icons and wait for the correct input (3.2).
- **Move list:** In the pause menu and Training; every technique with an input, its icons and points (section 9).
- **Input display:** Shows the player's stick motions and buttons as icons in Training, so a failed input is visible.
- **Loading tips:** One rule or technique tip per loading screen.
- **Coach feedback:** Drills recommended from recent match statistics.

7.8 Game Options

Option	Settings
Audio	Master, effects, referee voice, fighter and coach voice, crowd and music volume, each 0–100
Video	Resolution, window mode, quality preset (low, medium, high, epic), frame-rate cap (30, 60, 120, unlimited), V-sync
Controls	Vibration on or off; Skill Stick dead zone; stick input window 0.20–0.40 s (default 0.25 s); the full controls reference opens from here (7.10)
Saving	Settings and created fighters save automatically when changed
Career saving	Autosave after every event and at the end of every week; three career slots
Remapping and hold-to-toggle **[Future]**	See 7.9

7.9 Accessibility

The game is English only: all text, menus and voice-over. Korean referee commands and rank names are kept as the sport's own terminology, always with English on screen.

- **English subtitles for the Korean referee commands.**
- **Sides shown by position and label, not by colour alone: Chung on the left, Hong on the right, both labelled in text.**
- **Full button remapping, and hold-to-toggle for the four held modifiers (7.4)** **[Future]**.
- **Reduced camera motion and screen-shake options** **[Future]**.
- **Audio cues for incoming attacks and for stamina state** **[Future]**.

7.10 Controls Menu

The pause menu's move list (7.7) and Options (7.8) both draw from the reference below, split into the tabs a player would flip through without leaving the pause screen. Modifiers (RT, RB, LB, Y, LT) work exactly as in section 9; every technique row already spells out which ones apply, so this screen needs no cross-referencing during a match. Icons for every input are in 7.1, 7.2, 3.5 and section 9; this is the exhaustive text reference the Controls tab is built from. The tabs are Movement and Guards, Modifiers, Standing Kicks, Spinning & Jump Kicks, Fakes, Punches & Defense, and Referee Mode [Future]; Confirm, Back and Pause work the same on every screen (7.5) and have no tab of their own. On screen, a controller diagram lights up the buttons the selected row uses, and quarter, half and full circles are drawn as stick icons: a dot marks the start and an arrow shows how far to rotate. A quarter circle runs from 3 o'clock to 12 o'clock on the right side and mirrors on the left.

Movement & Guards

Action	Input	Notation
Movement	Left stick	Left stick
Switch Sides	A	A
Dash / Clinch	B	B; a dash that closes to contact range starts a clinch (11.3)
High Guard	D-pad Up	D-pad Up
Cross Guard	D-pad Left	D-pad Left
Mixed Guard	D-pad Right	D-pad Right
Low Guard	D-pad Down	D-pad Down — a normal guard, never a foul (8.4)

Modifiers (quick reference)

Modifier	Input	Effect
Back leg	Hold RT	Throws the technique with the back leg instead of the lead leg
Head	Hold RB	Aims the technique at the head instead of the trunk
Fake	Hold LB	Held with no stick motion it feints without a specific technique (2.3); held through a kick's input it fakes that kick (Fakes, 7.4, 8.6)
Jump	Tap Y before a kick's stick input	Throws that kick, or the Tornado Kick, airborne instead of grounded (7.4, 9.1, 9.2)
Active Defense	Hold LT	Reads the Skill Stick as defense instead of offense (8.8)
Standing vs. sliding	Left stick neutral vs. held	Neutral keeps the technique rooted in place; holding a direction slides or steps the fighter that way during it (7.4)

Standing Kicks

Technique	Input	Target · points
Side kick / cut kick	RS Up	Body · 2 / Head · 3; RB for the head
Side kick, back leg	Hold RT + RS Up	Body · 2 / Head · 3; RB for the head
Roundhouse kick	RS Left/Right, toward the lead leg	Body · 2
Roundhouse kick, back leg	Hold RT + RS Left/Right, toward the back leg	Body · 2 / Head · 3; RB for the head

Technique	Input	Target · points
Roundhouse kick, head	Hold RB + RS Left/Right, toward the lead leg	Head · 3
Hook kick	RS Left/Right, toward the side opposite the lead leg	Body · 2 / Head · 3; RT mirrors it to the lead leg's side for the back leg; RB for the head
Axe kick	RS Left/Right, forward quarter circle (3 o'clock to 12 o'clock), Down	Head · 3

Spinning & Jump Kicks

Technique	Input	Target · points
Spin side kick	RS Down, Up	Body · 4 / Head · 6; RB for the head
Spin hook kick	Hold RT + RS Left/Right Half Circle Down	Body · 4 / Head · 6; RB for the head
Tornado kick	Y, RS Left/Right Half Circle Down, Half Circle Up	Body · 4 / Head · 6; RB for the head
Jump side kick	Y, RS Up	Body · 2 / Head · 3; RT for the back leg; RB for the head
Jump roundhouse kick	Y, RS Left/Right, toward the lead leg	Body · 2 / Head · 3; RT mirrors it to the back leg; RB for the head
Jump hook kick	Y, RS Left/Right, toward the side opposite the lead leg	Body · 2 / Head · 3; RT mirrors it to the lead leg's side; RB for the head
Jump axe kick	Y, RS Left/Right, forward quarter circle (3 o'clock to 12 o'clock), Down	Head · 3
Jump spin hook kick	Y, Hold RT + RS Left/Right Half Circle Down	Body · 4 / Head · 6; RB for the head

Fakes

Fake	Input	Result
Fake, no kick	Hold LB, no stick motion	Feints without committing to a specific technique (2.3)
Fake side kick	Hold LB + RS Up	Startup plays, no active phase, no score (7.4, 8.6)
Fake roundhouse kick	Hold LB + RS Left/Right, toward the lead leg	Startup plays, no active phase, no score (7.4, 8.6)
Fake hook kick	Hold LB + RS Left/Right, toward the side opposite the lead leg	Startup plays, no active phase, no score (7.4, 8.6)

Fake	Input	Result
Fake axe kick	Hold LB + RS Left/Right, forward quarter circle (3 o'clock to 12 o'clock), Down	Startup plays, no active phase, no score (7.4, 8.6)
Fake spin side kick	Hold LB + RS Down, Up	Startup plays, no active phase, no score (7.4, 8.6)
Fake spin hook kick	Hold LB + RT + RS Left/Right Half Circle Down	Startup plays, no active phase, no score (7.4, 8.6)
Fake tornado kick	Y, Hold LB + RS Left/Right Half Circle Down, Half Circle Up	Startup plays, no active phase, no score (7.4, 8.6)
Fake a jump kick	Y, Hold LB + the kick's RS input	Startup plays, no active phase, no score (7.4, 8.6); RT and RB apply as on the real kick

Punches & Defense

Action	Input	Notation
Punch	X	Trunk · 1
Active Defense: high block	Hold LT, RS Up	Stops head-height attacks (8.8)
Active Defense: body block	Hold LT	Blocks punches and body kicks by closing the guard (8.8)
Active Defense: leg check	Hold LT, RS Down	Stops a body kick by lifting the leg; risks the leg-lifting penalty if held too long (8.4, 8.8)
Active Defense: sway / step	Hold LT, RS Left/Right	Moves out of range without a block (8.8)

Referee Mode [Future]

Its own tab, since these controls only apply in Referee Mode (3.5) and never overlap with a fighter's inputs above.

Action	Input	Notation
Move on the mat	Left stick	Left stick
Turn and look	Right stick	Right stick
Select Chung	LB	LB
Select Hong	RB	RB
SHI-JAK (start) or KYE-SOK (continue)	A	A
KAL-YEO (break)	B	B
GAM-JEOM to the selected fighter	Hold X, choose the reason with RS, release	Hold X, choose with RS, release
KEU-MAN (stop)	Y	Y

Action	Input	Notation
Knockdown count	Hold LT	Hold LT
Raise the selected fighter’s hand	RT, after the final KEU-MAN	RT
CHARYEOT and KYEONGRYE	D-pad Up	D-pad Up
JOON-BI (ready)	D-pad Down	D-pad Down
Pause	Menu button	Menu button

8 Core Sparring Systems

The systems that make a sparring match: the systems v1.0 needs. The technique list is section 9 and the data behind each technique is 16.3.

8.1 Systems Overview

System	Intent	Type
Match format (8.2)	Matches play like real WT sparring, so practitioners recognize them and tactics transfer	Player-facing
Scoring (8.3)	Harder, riskier techniques earn more points, as in the sport	Player-facing
Hit detection	Only clean, in-range contact during the active phase scores, so distance matters	Internal
Penalties (8.4)	Stops boundary running, stalling and illegal contact	Player-facing
Referee (8.5, 11.2)	Frames the match with the sport's etiquette and commands	Player-facing
Skill Stick (8.6)	Kick choice is a readable stick gesture; held modifiers choose leg and target	Player-facing
Input recognition and buffer	Turns stick gestures and modifier holds into techniques within a 0.25 s window	Internal
Stance and guards (8.7)	Lead and back legs create open and closed stance matchups; guards change posture	Player-facing
Active defense (8.8)	A held defensive state for answering attacks	Player-facing
Stamina (8.9)	Limits repeated high-commitment techniques without locking controls	Player-facing
Combat state machine	Startup, active and recovery phases create counter windows	Internal
Hit reactions (8.10)	Hits read clearly while the match keeps flowing	Player-facing
AI opponent (11.1)	A sparring partner that uses real tactics	Internal
Video review (8.11) **[Future]**	Players challenge a missed call, as coaches do	Player-facing

8.2 Match Format

The default ruleset follows World Taekwondo (WT) sparring. Values here are the game's reference; confirm them against the current WT Competition Rules before locking a release, and update them when WT revises its rules.

Rule	Default	Options
Rounds	Best of 3	Single round, best of 3
Round length	2:00	1:00 · 1:30 · 2:00
Break between rounds	1:00	Skippable in local play

Rule	Default	Options
Round win	Most points at time, or point gap	Point gap on/off
Point gap	12 points	Off · 12 · 20
Contest area	Octagonal mat with boundary line	Arena choice
Sides	Chung blue and Hong red	Assigned per match, never chosen as a colour

8.3 Scoring

Hits register the way the electronic protector and scoring system (PSS) does: the trunk protector needs a valid kick above an impact threshold or a valid punch, and the head needs valid kick contact. In game, a hit registers when the attacking limb's trace reaches the target zone during the active phase with enough impact.

Technique	Target	Points
Punch	Trunk protector	1
Kick	Trunk protector	2
Spinning kick	Trunk protector	4
Kick	Head	3
Spinning kick	Head	6
Opponent penalty (감점 gam-jeom)	—	+1

Tie-breaks follow WT order (higher-value techniques first, then referee decision). Verify the current tie-break order and per-round penalty limit.

8.4 Penalties

A gam-jeom gives the opponent one point, and a fighter who reaches 5 gam-jeom in a round loses it (starting value; verify the current WT per-round limit, 18.1). Penalties the game detects, and exactly what triggers each one:

- Boundary: both of the fighter's feet are completely outside the mat's edge (6.3) at the same instant. One foot outside, or a foot only touching the line, does not trigger it.
- Falling: the fighter enters the Knocked Down combat state (16.3) from any trigger other than the opponent's landed technique. See 8.10 for exactly how the game tells a foul fall apart from a real knockdown.
- Avoiding or delaying: the fighter's forward facing points more than 90° away from the opponent for more than 1.0 s (turning the back); the fighter's stance height drops below Low Guard's own posture (7.3) — lower than any of the four guards allow — for more than 1.0 s without attacking (crouching, never triggered by Low Guard itself, a normal

guard choice); or neither fighter's technique reaches its active phase for 6 s straight, the passivity timer (not engaging) — starting values; verify against current WT rules (18.1).

- Leg lifting: the lead or back leg held above hip height for more than 3 continuous seconds while outside that leg's own technique (Attack Active or Attack Recovery, 16.3).
- Kicking below the waist: a kick's Attack Active trace (16.3) registers below the belt-line height on the target's collider (11.3). Striking the head with the hand: a punch's Attack Active trace registers on the head hitbox instead of the trunk — already unscorable (8.3), and a foul here too.
- Grabbing, holding or pushing: sustained contact between the fighters for more than 1.0 s while both are inside clinch range (11.3), outside either fighter's Attack Active phase.

8.5 Referee Commands

Command	Meaning	Command	Meaning
CHARYEOT 차렷	Attention, before the bow	**KAL-YEO** 갈려	Break, fighters separate
KYEONGRYE 경례	Bow	**KYE-SOK** 계속	Continue
JOON-BI 준비	Ready	**KEU-MAN** 그만	Stop, end of round
SHI-JAK 시작	Start, the clock runs	**GAM-JEOM** 감점	Penalty declared
HANA … YEODEOL 하나 … 여덟	Knockdown count, one to eight	**CHUNG / HONG** 청 / 홍	Side named with a gam-jeom and with the winner
CHUNG SEUNG / HONG SEUNG 청 승 / 홍 승	Winner, as the referee raises the winner's hand	**KYE-SHI** 계시 **[Future]**	Time-out: the clock stops for a video review or injury

8.6 Skill Stick

The right stick (R) is the Skill Stick. A direction or motion chooses the kick (7.2); held modifiers change it: RT kicks with the back leg, RB aims at the head and LB turns the technique into a fake: its startup plays, then the fighter recovers without reaching the active phase, no score and no commitment. Y adds a jump: tap it just before a side, roundhouse, hook, axe or spin hook kick's gesture and that kick throws airborne instead of grounded, same target and points, at the Jump cost tier (8.9). Y can't be held through the motion like RT, RB or LB, since the same right thumb drives the Skill Stick and can't hold a face button and steer the stick at once — so it has to be a tap, released before the gesture starts. The Tornado Kick uses that same tap; what makes it the hardest technique in the game is the motion that follows — a full circle, starting downward, swept in one clean motion (9.2). Holding LT instead reads the Skill Stick as Active Defense, not offense (8.8). Every technique with an input is listed in section 9.

Animation flow for every kick (captured, then driven by gameplay):

BOUNCE → CHAMBER → TECHNIQUE → RECHAMBER → LAND → BOUNCE

8.7 Stance and Guards

- **Stances:** Left Lead and Right Lead. Switch Sides (A) plays a visible switch animation; the fighter is never instantly mirrored. Open and closed stance matchups emerge from positioning.
- **Guards** differ in posture, readiness and transition timing, not stat bonuses. All four guards ship in v1.0; their inputs are in 7.3, and each one is described below.
- **High Guard**: hands high and upright. The quickest guard to throw a head kick from and the slowest to start a body kick from.
- **Cross Guard**: rear hand forward across the body. A quick guard that covers body kicks on the open side.
- **Mixed Guard**: one hand high and one low. The balanced guard, the starting guard, and the quickest to change to any other guard.
- **Low Guard**: compact and lowered. The quickest guard to start a body kick from and the slowest to defend a head kick from. It is a normal guard and never counts as crouching (8.4).
- **Timing**: changing guard takes 0.15 to 0.25 s, with no stat bonus for any guard. These are starting values, tuned in playtests (18.1).

8.8 Active Defense

Holding LT puts the fighter in Active Defense Mode: the Skill Stick reads as defense instead of offense. Holding LT alone closes the guard with the hands and arms, blocking punches and body kicks; RS Up instead reads as a high guard block, stopping head-height attacks; RS Down is a leg check, lifting the leg to block a body kick instead — held too long it risks the leg-lifting penalty (8.4), the same trade-off a real leg check makes; RS Left/Right is a sway or step off the line, moving out of range without committing to a block. Releasing LT returns the Skill Stick to offense. Active Defense costs stamina like an attack and has its own recovery, so a mistimed block or sway still leaves an opening (8.9, tuned in playtests).

8.9 Stamina

Stamina is short-term explosiveness. Low stamina lengthens recovery and reduces explosiveness; it never locks out controls.

Action or state	Effect	Starting value
Stamina pool	—	100 (Equal Terms; the Stamina attribute changes it in Career)
Normal footwork and bouncing	No cost	0
Punch 	Small cost	3
Basic kick	Small cost	8
Dash 	Moderate cost	12
Jump or spinning kick	High cost	20
Tornado kick	Highest cost	30
Low stamina	Longer recovery, less explosive techniques, heavier breathing (1.10)	Below 25: recovery phases 30% longer, impact 20% lower
Regeneration	Starts after low activity	15 per second after 1.0 s without spending stamina
Round break	Refill	Back to 100

Starting values for the Milestone 2 prototype, tuned in playtests (18.1).

8.10 Hit Reactions

Severity	Response	Control lost
Minor	Additive torso or head reaction; movement continues	0 s
Normal	Partial or full-body reaction	0.3–0.5 s
Heavy	Captured stagger or loss of balance	0.8–1.2 s
Knockdown	Captured fall, grounded state and get-up	2.5–4.0 s
Physical falls **[Future]**	Animation-to-ragdoll blend for off-balance falls	As knockdown

What triggers Knocked Down. Two paths lead to the same Knocked Down state (16.3), and the Penalty system (8.4) tells them apart by which one fired. From a landed hit: a technique's impact (MoveData, 16.3), weighed against the defender's Composure and remaining stamina (5.4, 8.9), resolves to Knockdown severity (above), and Hit Reaction transitions straight into Knocked Down — the referee starts the count (8.5, 11.2), never a penalty. Without a landed hit: a high-commitment technique thrown from bad distance can fail its own landing, so its Attack Recovery transitions into Knocked Down directly instead of returning to Idle/Movement — a per-technique failure chance in MoveData (16.3, starting at 0 for grounded kicks, non-zero for spinning, jumping and the Tornado Kick), tuned in playtests. Any Knocked Down entry that did not come from a Hit Reaction is the Falling penalty (8.4).

8.11 Video Replay Request [Future]

Each corner has one video review per match. During a stoppage a player can challenge a missed score or a penalty; the game replays the exchange from scoring data. A successful challenge corrects the score and keeps the review.

8.12 Rulesets [Future]

- **Custom rules:** Round count and length, point values, point gap, penalty limits, saved as named presets and shareable online.
- **Historical WT rules:** Earlier scoring eras for exhibition matches.
- **Other styles:** Additional sparring formats (for example ITF-style continuous point sparring) as separate rulesets with their own gear and techniques, if licensing and research allow.

8.13 Epics and User Stories

v1.0 work breaks down into these epics. Each story is a testable requirement for its milestone (section 20).

Epic	User stories	Milestone
E1 Skill Stick input	As a player, I push the right stick up to throw a side kick, and hold RT to throw it with my back leg. As a player, I see my last input as controller icons on the input display, so I know why a kick did not come out.	M1
E2 Stance and legs	As a player, after I switch sides with A, holding RT still kicks with my new back leg.	M1
E3 Combat states and hit detection	As a player, a kick that reaches the protector during its active phase scores, and a kick thrown from out of range misses.	M1
E4 Motion capture pipeline	As the developer, I turn one captured roundhouse kick into a retargeted MetaHuman animation in Unreal, by whichever capture method Milestone 1 selects.	M1
E5 Match rules and scoring	As two players, we play a best-of-3 match with 2:00 rounds, WT point values, penalties and a 12-point gap.	M2
E6 Stamina	As a player, repeating spinning kicks drains my stamina and slows my recovery, but I never lose control.	M2
E7 Local versus	As two players with two controllers, we finish a full match on one console.	M2
E8 Production animation	As a player, techniques, transitions and reactions use production mocap and blend without visible pops.	M3
E9 Game flow and UI	As a player, I go from the title screen through setup, a match and results, and back, without restarting the game. As a solo player, I run a local tournament bracket to its final, with AI fighters filling every slot no player takes.	M4

Epic	User stories	Milestone
E10 AI opponent	As a solo player, I spar an AI that manages distance, fakes and counters at a difficulty I choose.	M4
E11 Training mode	As a new player, I finish the six Training lessons listed in 3.2.	M4
E12 Career season	As a career player, I plan a training week, enter a tournament, pass a grading exam, and see my rank, ranking points and newly unlocked techniques change.	M4
E13 Balance and polish	As a tester, my beta feedback changes MoveData tuning values without code changes.	M5

9 Techniques

Every technique with an input, its points and the input that performs it; techniques still waiting on an input are in 9.4. Points follow 8.3. Holding RB takes any kick to the head for the head point value (punches ignore RB and only score to the trunk), so each table gives the body value first and the head value where the technique can reach it.

9.1 Kicks

Technique	Target · points	Input	Notation
Side kick / cut kick	Body · 2 / Head · 3		RS Up; RB for the head
Side kick, back leg	Body · 2 / Head · 3	Hold  + 	RT + RS Up; RB for the head
Roundhouse kick	Body · 2		RS Left/Right, toward the lead leg (18.2)
Roundhouse kick, back leg	Body · 2 / Head · 3	Hold  + 	RT + RS Left/Right, toward the back leg — mirrored from the lead-leg version (18.2); RB for the head
Roundhouse kick, head	Head · 3	Hold  + 	RB + RS Left/Right, toward the lead leg
Hook kick	Body · 2 / Head · 3	 toward the side opposite the lead leg; RT mirrors it to the lead leg's side for the back leg (18.2)	RS Left/Right, toward the side opposite the lead leg; RB for the head
Axe kick	Head · 3	 then 	RS Left/Right Quarter Circle Forward, Down

9.2 Spinning and Jumping Kicks

Technique	Target · points	Input	Notation
Spin side kick	Body · 4 / Head · 6	 then 	RS Down, Up; RB for the head
Spin hook kick	Body · 4 / Head · 6	 + 	RT + RS Left/Right Half Circle Down; RB for the head
Tornado kick	Body · 4 / Head · 6	 , then  then 	Y, then RS Half Circle Down, then Half Circle Up; RB for the head
Jump side kick	Body · 2 / Head · 3	 , then 	Y, then RS Up; RT for the back leg; RB for the head

Technique	Target · points	Input	Notation
Jump roundhouse kick	Body · 2 / Head · 3	Y, then RS Left/Right	Y, then RS Left/Right, toward the lead leg; RT mirrors it to the back leg; RB for the head
Jump hook kick	Body · 2 / Head · 3	Y, then RS Left/Right	Y, then RS Left/Right, toward the side opposite the lead leg; RT mirrors it to the lead leg's side; RB for the head
Jump axe kick	Head · 3	Y, then RS Left/Right Quarter Circle, then RS Down	Y, then RS Left/Right Quarter Circle Forward, Down
Jump spin hook kick	Body · 4 / Head · 6	Y, then RT + RS Left/Right Half Circle	Y, then RT + RS Left/Right Half Circle Down; RB for the head

Spinning kicks score the higher trunk and head values (4 and 6) when they register as spinning techniques.

Jump kicks throw airborne instead of grounded, at the same target and points as the grounded version, and cost stamina at the Jump tier (8.9). The Tornado Kick uses the same tap of Y as every other jump kick; only its motion differs — a full circle starting downward, deliberately the hardest gesture in the game to sweep cleanly in one motion (18.2).

9.3 Punches, Movement and Other Actions

Action	Target · points	Input	Notation
Punch	Trunk · 1	X	X
Fake attack	—	Hold LB	Hold LB
Active defense	—	Hold LT	Hold LT
Dash	—	B	B
Switch sides	—	A	A
Movement	—	LS (assumed)	Left stick (assumed)
Switch guard	—	D-pad: High, Cross, Mixed, Low	D-pad: Up high, Left cross, Right mixed, Down low

Punches only score to the trunk protector; punches to the head are penalized.

9.4 Techniques Not Yet Mapped

These techniques have no input assigned yet: push kick, front kick, fast lead-leg roundhouse, double and triple kicks, 360° and jumping back kicks, close-range kicks in the clinch and clinch actions (arm blocks, sway and the leg check are covered by Active Defense, 8.8). They stay out of the move list until they have inputs (18.1); candidates for a larger move library are in section 17. Jump variants of the side, roundhouse, hook, axe and spin hook kicks are in the move list (9.1, 9.2); only jump variants of the techniques above (9.4) stay cut.

10 Customization

Fighters are MetaHuman bodies wearing the dobok and sparring gear, refit to each body. Gear is a removable layer: matches show full gear; menus and Career show the fighter in dobok. The fighter creator is part of Career; Sparring and Tournament use premade men's and women's fighters (7.5).

10.1 Options

Category	Options	Rule
Sex	Male or female	Chooses the male or female MetaHuman body set; sets the roster the fighter appears in; it has no effect on rules, scoring or ratings (5.4.2)
Body	Five MetaHuman body types, each with a set height and weight (5.4.2)	Sets reach, build and Career base ratings
Fighting style	Five styles (5.4.3)	Career base ratings; changeable once per season
Face	MetaHuman face presets and blends	—
Skin	MetaHuman skin-tone chart	Real skin tones only
Hair, brows, facial hair	MetaHuman grooms	Natural pigment only; no dye
Eyes	Iris colour per eye, pattern	Natural eye colours only; no tints
Side colour	Chung or Hong	Assigned by the match (10.2)
Belt	Rank colour	Earned rank
Dobok collar	White, black; red-black for poom	Follows rank
Dobok back	Name and country code	As printed on competition doboks
Defaults	Preferred lead leg, starting guard	Starting state only
Gear models **[Future]**	Helmet, protector, glove and foot-guard styles	Visual only; side colours still apply
Dojang **[Future]**	Club space, banners, club emblem	Emblems moderated online
Ceremony **[Future]**	Entrance walk, podium celebration, respectful gestures	No taunts

10.2 Side Colours

The chest protector and helmet are only ever Chung blue or Hong red, and the match assigns the side. There is no colour picker for gear, in any phase. Hong red is #C8102E; Chung blue is the protector's baked texture colour.

10.3 Natural Appearance

- **Skin:** MetaHuman skin-tone chart only; no free tint.
- **Hair:** Melanin, redness, greying and lightness only; dye colours stay off.
- **Eyes:** natural iris presets (dark brown, brown, amber, hazel, green, blue, grey), set per eye; no tints.

10.4 Unlock Sources

- Rank: belt colours, dan collar, signature technique slot.
- Career milestones and challenges: podium poses, medals and gear styles (10.1) **[Future]**.
- Any future paid content is cosmetic only, never ranks, techniques or attributes **[Future]**.

11 Game Objects and AI

11.1 AI Opponents

Feature	Description
Difficulty levels	Reaction time, read accuracy, stamina management and technique variety scale by level
Uses real tactics	Manages distance, fakes, guards, counters and the boundary instead of reading inputs
Fighter styles	The fighting styles in 5.4.3: all-rounder, counter fighter, pressure fighter, head-kick specialist, cut-kick blocker
Rivals	Named Career opponents who can switch fighting style between seasons, like the player (5.4.3)
Adaptive AI **[Future]**	Learns a player's habits within a match (favourite kick, stance, timing) and adjusts
Shadow fighters **[Future]**	AI trained from a player's own matches that friends can spar against offline

11.2 Officials

The referee is a non-playable character on the mat. Six referees built from MetaHuman presets (Kelvin, Bo, Jorge, Omari, Walter and Vivian) wear a light-blue shirt, navy tie, beige trousers, brown belt and white sneakers (1.10). Each match picks one at random, and each referee has their own voice (1.10). The referee speaks only the commands in 8.5. In Referee Mode the player controls the referee instead **[Future]** (3.5).

Situation	Referee behaviour
Match start	Calls charyeot, kyeongrye, joon-bi and shi-jak from between the fighters, then steps back
During play	Moves to keep both fighters in view without blocking the camera
Clinch	Calls kal-yeo, steps between the fighters, then calls kye-sok
Penalty	Stops play with kal-yeo, names the penalized side and declares gam-jeom (for example Hong, gam-jeom), including for stalling (8.4), then calls kye-sok
Knockdown	Counts hana to yeodeol (one to eight) while the fighter gets up, then calls kye-sok
Video review or injury **[Future]**	Calls kal-yeo and kye-shi to stop the clock, then kye-sok when play resumes (8.11)
End of round and match	Calls keu-man; at match end raises the winner's hand and calls Chung seung or Hong seung

11.3 Equipment and Props

Object	Attributes and values	Behaviour
Trunk protector (PSS)	Side colour Chung or Hong; impact threshold (value unresolved, 18.1)	Registers a kick above the threshold, or a punch, as a score (8.3)
Helmet	Side colour Chung or Hong	Registers valid kick contact to the head
Foot guards and gloves	White	Carry the limb traces that score
Mat and boundary	8 m octagon in a 12 m square	Boundary check on both feet (8.4)
Arena scoreboards	Up to four screens; one portable scoreboard at the club dojang (6.2)	Mirror the HUD scoreboard
Kicking paddles and reaction targets	Height and timing set per drill	Light up to show the target; grade speed and accuracy
Training dummy	Stands still or loops a recorded movement	Free practice target

Technique and object values live in a MoveData data table (16.3) that exports to CSV for review. The table does not exist yet (18.1).

12 Online and Community [Future]

Feature	Description
Rollback netcode	Responsive 1v1 online play; requires deterministic combat simulation (16.3)
Casual and private rooms	Quick match, invite codes, rematch
Ranked seasons	Skill rating; season rank shown on a nameplate, never as a belt
Fair play	Disconnect penalties, report tools, name and emblem moderation
Clubs	Online dojangs with members, internal ladders and club-vs-club team events
Community tournaments	Organizer tools, brackets, check-in, broadcast mode (3.4)
Cross-play	PC and consoles in one player pool

13 Presentation

- **Match intro: fighters enter in full gear, then the referee's opening commands (11.2).**
- **Camera:** Side-on combat camera that keeps both fighters and the boundary in frame; 1.5 s cuts on spinning-kick scores.
- **Menus:** The player's fighter out of gear in dobok, in a dojang or locker-room setting.
- **Arenas: the six Career venues, one for each tournament tier, listed in 6.2.**
- **Commentary** **[Future]**: play-by-play and colour commentary.

14 Localization

- **Plan:** English only, in every phase. No other language is planned.
- **Korean terminology:** Referee commands, rank names and technique names appear in romanization with English meaning, for example KAL-YEO 갈려: Break. They are the sport's terms, not translations.
- **String syntax:** All player-facing text is FText stored in Unreal String Tables, never hard-coded in Blueprints or C++. Values inside strings use FText::Format arguments, for example {Points}.
- **String tables:** ST_UI, ST_Techniques, ST_Referee and ST_Career, in Content/Localization/StringTables (planned; the folder does not exist yet).
- **Key names:** Area.Group.Item, for example UI.MainMenu.Play, Technique.RoundhouseKick.Name, Referee.KalYeo.Subtitle.
- **External files:** None; String Tables export to CSV for proofreading.

15 Tools

Tool	Purpose	Location
MetaHuman Creator	Fighter and referee bodies, faces, skin, hair and eye presets	Unreal Engine 5.8 plugin
Blender pipeline scripts	Rig, split, mask and refit the dobok and gear onto MetaHuman bodies; bake hidden face maps; build referee outfits and environment meshes	Tools/Blender
Unreal Python scripts	Create fighters and referees, import gear as wardrobe items, set up Chaos cloth, run range-of-motion tests, build the arena and dojang levels	Tools/Unreal
Environment texture generator	Mat and venue textures	Tools/Environment
OpenStanceEditorTools	C++ editor module that exposes cloth creation to Python	Source/OpenStanceEditorTools
Motion capture solver	Turns recorded performance into skeletal animation. Neither the capture method nor the solver is chosen; Milestone 1 compares them	External
IK Rig and IK Retargeter	Retarget mocap onto the MetaHuman skeleton	Unreal Engine
MoveData table	Technique timing, stamina cost, points, target and range, editable without code	Unreal Engine (planned)
Combat debug overlay	Input icons, leg, move, target, stamina, hit result and score in play	In game (planned)
Perforce Helix Core and P4V	Version control for the Unreal project, with binary asset locking	AWS EC2

16 Technical Design

16.1 Stack

Area	Choice
Engine	Unreal Engine 5.8 (source build), OpenStance.uproject
Characters	MetaHuman Creator; dobok and gear as outfit pieces refit per body (Tools/Blender)
Skeleton	MetaHuman body skeleton; gear follows with Leader Pose; mocap retargets with the IK Retargeter
Motion capture	Method not yet chosen. Markerless video capture (MetaHuman Animator, Rokoko Vision) and suit-based capture are both candidates; the Milestone 1 feasibility test decides (18.1, section 19)
Version control	Perforce Helix Core on AWS for the project; Git for tools, docs and portfolio

16.2 Animation Architecture

Need	Approach
Continuous movement	Animation Blueprint and Blend Spaces
Committed attacks	Animation Montages
Hit windows	Anim Notifies driving limb traces and scoring windows
Guards	Layered Blend Per Bone
Mocap retargeting	IK Rig and IK Retargeter
Foot planting	IK correction
Range alignment	Limited Motion Warping
Minor reactions	Additive or layered reactions

16.3 Data and Simulation

- **MoveData:** Animation, input gesture and modifiers, startup / active / recovery timing, stamina cost, point value, target, range, reaction properties, and a self-fall chance on Attack Recovery for high-commitment techniques (8.10). Separate from animation assets so placeholders swap for final mocap without changing combat logic.
- **Fighter runtime state:** Stance, guard, lead and back leg, held modifiers (active defense, back leg, fake, head), a jump-primed flag from a leading tap of Y, combat state, stamina, score, match state.
- **Combat states:** Idle/Movement, Attack Startup, Attack Active, Attack Recovery, Active Defense, Hit Reaction, Staggered, Knocked Down, Recovering, Match Finished.
- **Fighter profile:** Appearance, body type and fighting style; level, XP, rank, attributes, unlocked techniques and career save.

- **Deterministic combat step** [Future]: fixed-tick combat simulation separated from rendering and animation, so rollback netcode and input-based replays are possible. Combat code in v1.0 is structured so this refactor stays small.
- **Services** [Future]: accounts, matchmaking, ranked seasons, club and tournament backend.

16.4 Major Technical Risks

Risk	Mitigation
Capture struggles with fast kicks and spins	Early feasibility test on a roundhouse kick and a spin hook kick, comparing capture methods and solvers before production (section 19). If spinning kicks are still not clean by Milestone 1, the December build (Milestone 2) is built on the grounded kicks and whatever the lab session captured cleanly. Any spinning or jump kick without clean capture ships with a temporary placeholder: the matching grounded technique with a boost in height. Production mocap replaces each placeholder as soon as the capture or solver issue is fixed (20.1).
Animation workload grows too large	Shared transitions, layering, mirroring and a tightly scoped move list. Jump kicks (9.2) reuse their grounded kick's captured technique with a shared launch-and-land layer blended underneath (16.2), rather than a bespoke capture per jump kick; only the Tornado Kick is a fully unique capture
Mocap looks authentic but attacks feel unresponsive	Gameplay timing separate from performance timing: montages, notifies, limited blending and warping
Physics and ragdoll take too long	Authored mocap falls in v1.0; animation-to-ragdoll blending is Future work (8.10)
Skill Stick gestures are hard to perform or misread	Prototype gesture recognition first, with an input-icon debug display and practitioner playtests. The Tornado Kick's full circle, starting downward, is deliberately the hardest gesture in the game to sweep cleanly (7.4, 9.2); tune how forgiving that recognition is in playtests
Scope grows beyond v1.0	Online play, online tournaments, challenges, replays, video review and a large roster are Future work (section 17)
A console release needs certification	The game targets PlayStation 5 and Xbox Series X|S but is built on Windows, and developer accounts, devkits and certification are not in place (18.1). The First Half milestones (M1, M2; ending December 2026) demo on Windows only; PlayStation 5 and Xbox Series X|S remain the v1.0 target for the finished capstone in Spring 2027, once developer accounts and devkits are in place. Nothing in the design depends on a console-only feature.
Career is a second mode for one developer	Career reuses the sparring core rather than adding systems: the same match, AI, contest area and scoring, with a calendar, grading and brackets on top. Its events and venues are data and set dressing, so it scales down to fewer events without code changes, and any venue not built falls back to the competition arena (sections 4, 5 and 6.2).

16.5 Project Structure and Naming

Path	Contents
Content/Audio	Voice, effects, ambience and music (planned)
Content/Characters/Fighters	Base fighter, gear wardrobe items, hidden face maps
Content/Characters/Officials	Referees and their outfits
Content/Environment	Materials, meshes and textures
Content/Maps	L_CompetitionArena, L_TrainingDojang; the Career venues in 6.2 are planned
Source/OpenStance	Runtime game module: combat, match, input, AI
Source/OpenStanceEditorTools	Editor-only C++ helpers
Tools/Blender, Tools/Unreal, Tools/Environment	Pipeline scripts (section 15)
Resources	Source art: models, references, textures
Docs	Design documentation

Asset prefixes follow the Unreal conventions already used in the project: L_ levels, MH_ MetaHumans, SK_ skeletal meshes, M_ materials, MI_ material instances, T_ textures, WI_ wardrobe items. Planned: DT_ data tables, AM_ animation montages, ABP_ animation blueprints, BS_ blend spaces, ST_ string tables, SW_ sound waves, MS_ MetaSound sources.

16.6 Data Formats

- **MoveData:** An Unreal Data Table of a C++ row struct, one row per technique; exports to CSV.
- **Saves:** Settings and fighter profiles as Unreal SaveGame objects; Career slots use the same format.
- **Animation and gear imports:** FBX in centimetres (UnitScaleFactor 1), matching the MetaHuman export.
- **Replays [Future]: input** and scoring-event streams replayed by the deterministic combat step (16.3).

16.7 AI and Autonomous Systems

The AI opponent (11.1) sends the same inputs as a player and uses the same MoveData; it never reads the player's inputs. Planned structure: a State Tree picks a tactic (keep distance, pressure, fake, counter) and a utility score picks a technique from distance, stamina, score and clock. Difficulty scales reaction time from 450 ms at the easiest level to 180 ms at the hardest (starting

values), read accuracy, stamina management and technique variety. The game needs no procedural world, no pathfinding beyond the mat and no economy.

16.8 Test Cheats

Unreal console commands, in development builds only.

Command	Effect
os.InfiniteStamina 0|1	Stamina never drains
os.SetScore <Chung> <Hong>	Sets the score
os.SetClock <seconds>	Sets the round clock
os.AIMode idle|block|mirror|normal	Freezes or scripts the AI
os.ShowHitboxes 0|1	Draws limb traces and target zones
os.CombatDebug 0|1	Shows the combat debug overlay with input icons
slomo <scale>	Unreal time dilation, for example slomo 0.25
os.SetRank <rank>	Sets Career rank and its unlocks

17 Ideas and Expansions

Everything beyond v1.0. Throughout this document a **[Future]** tag marks a deferred feature where it belongs in a system, so each system reads as a whole; this section summarizes them. None of it is required for v1.0: it is design kept on file in case the main features are finished ahead of schedule.

v1.0: What this capstone delivers	Future: Every sparring format
Sparring, Training and local Tournament Career: calendar, training, grading, career events Levelling, ranks, attributes, technique unlocks AI opponent, competition arena and training dojang Career venues for event tiers 1 to 6	Online casual and ranked, rollback netcode Challenges and Replays Video review; animation-to-ragdoll falls Larger move library, including the techniques in 9.4 Accessibility options: remapping, reduced motion, audio cues Team Championship, para gyeorugi, cadet and junior careers Clubs, community tournaments, broadcast Adaptive and shadow AI Custom and historical rulesets (8.12) Commentary (13) Gear styles, dojang and ceremony customization (10.1) Ageing, retirement, hall of fame and the world games (4.2, 4.4) Referee Mode with its own controls (3.5) PC release and cross-play

17.1 Later Platforms

- **PC release on Steam, with keyboard play and full remapping.**
- **Cross-play across PlayStation, Xbox and PC.**

18 Unresolved Questions

Open questions are listed here. When one is resolved it moves to 18.2 with a link to the section that answers it; it is never deleted.

18.1 Open

Controls

- Inputs for the techniques in 9.4, or whether they are cut.
- Playtest the Skill Stick gestures and the 0.25 s gesture window.

Rules and tuning

- Current WT tie-break order and per-round penalty limit. The WT Competition Rules and Interpretation in force from 1 June 2026 have not been read in full; sources disagree on the limit (4 gam-jeom in the 2022 interpretation, 5 here).
- Verify WT values: start-mark spacing, the 3-second leg-lift limit, passivity timer length, the trunk protector impact threshold, and the referee's side, winner and time-out calls (8.5). Also check the passivity penalty in the last 10 seconds of a round (sources give both 2 points to the opponent and 3 gam-jeom) and the boundary foul (any part of the foot over the line in the 2026 rules, against both feet here, 8.4).
- Number of AI difficulty levels and their tuning values (16.7); AI complexity for v1.0.
- Tune starting values after Milestone 2 playtests: movement (6.4), stamina (8.9), hit reactions (8.10).
- Final move count for v1.0.
- Referee Mode **[Future]**: tune the referee rating values and the warning (40) and removal (20) thresholds (3.5.1).
- Whether the attributes taller body types give up balance their reach in Career (5.4.2), and whether reach needs its own balancing under Equal Terms, where body type modifiers do not apply (5.6, 10.1).
- Tune the body type and fighting style modifiers, the starting OVR of 64, the attribute caps, and how many hours of Career play it takes to reach OVR 99 (5.4).

Production

- Motion capture method and solver: markerless video capture (MetaHuman Animator, Rokoko Vision) or a suit-based system. Milestone 1 decides (section 19).
- Amount of Motion Warping.
- Natural eye-colour preset values (iris chart sampling in the editor).
- Wireframe mockups for every screen (7.5): first pass done as a clickable prototype and a Lucid wireframe sheet; needs review.

- Create the art asset spreadsheet (1.10) and the MoveData table (16.3).
- Music: commissioned or licensed (1.10).
- Voice casting: six referee voices, at least two men's and two women's fighter voice sets and two coach voices, with every Korean read checked by a Korean-speaking referee or instructor (1.10).
- Per-platform performance targets, in resolution and frame rate, measured on the Milestone 3 build (1.11).
- Second Half dates once BCIT publishes the 2027 schedule.
- Console developer accounts, devkits and certification for PlayStation 5 and Xbox Series X|S; v1.0 runs on Windows (1.11).
- Trophies and achievements: confirm the grade mix, the Gamerscore split and the icon sizes against Sony's and Microsoft's current certification requirements once developer accounts exist (5.7).
- Legal check of every pop culture reference in trophy names and easter eggs (5.7, 6.5); rename any that Sony, Microsoft or a rights holder objects to.
- Belt colour order for release; Career event names.
- Online service provider and the price of a commercial release **[Future]**.
- Licensing path for any real federation branding or events; research partners for para taekwondo [Future].

18.2 Resolved

Question	Resolution	See
Controller layout	Fixed on 2026-09-14	Section 7
Skill Stick direction	RS toward the side of the leg throwing the kick is the roundhouse, the other side is the hook; holding RT for the back leg mirrors both, since the back leg is on the opposite side from the lead leg (8.7) — this follows the fighter's own stance, not a fixed screen direction	7.2, 8.7
Active Defense	Holding LT reads the Skill Stick as defense: held alone it blocks punches and body kicks, RS Up blocks high, RS Down is a leg check, RS Left/Right sways or steps off the line	8.8
Tornado Kick input	Taps Y like every other jump kick, instead of holding LB through the motion like a fake attack, so it doesn't ask the right thumb to hold Y and steer the Skill Stick at once. What sets it apart from a Jump Spin Hook Kick is the motion itself, not the tap: a full circle starting downward (9.1, 9.2), deliberately the hardest gesture in the game to sweep cleanly	7.2, 9.2
Engine	Unreal Engine 5.8 source build	16.1
Version control	Perforce Helix Core on AWS; Git for tools, docs and portfolio	16.1
Scope of the sport	Sparring only: no poomsae, breaking or club-management mode	Section 1

Question	Resolution	See
Gear colours	Protector and helmet are only Chung blue or Hong red, assigned by the match	10.2
Appearance	Natural skin, hair and eye colours only	10.3
Character skeleton	MetaHuman body skeleton; gear follows with Leader Pose	16.1
Contest area size	8 m octagon in a 12 m square	6.3
Ragdoll	Authored mocap falls in v1.0; ragdoll blending is Future	8.10
Languages	English only	Section 14
Target platform	Console-first for the finished capstone (Spring 2027): PlayStation 5 and Xbox Series X|S. The First Half milestones (M1, M2; ending December 2026) are developed and demonstrated on Windows only	1.11
Phases	Two: v1.0 is what this capstone delivers, Future is after it	Main page
Weight classes and divisions	Cut for playability: one open division, no gender-based rules or division. The Men or Women choice only picks the premade roster and body set, and either roster can face the other. Body types keep a height and weight, which set reach, build and Career base ratings only (5.4.2, 7.5)	5.4.2, 7.5, 10.1
Kicks without RT	Use the lead leg; RT selects the back leg	7.4, 8.6
Spin hook kick and RT	RT is required: the spin hook is thrown by the back leg, so its gesture is mirrored like any back-leg kick	7.2, 9.2
Guard postures	High, Cross, Mixed and Low differ in posture, readiness and transition timing with no stat bonuses; Mixed is the starting guard and Low is never crouching	8.7, 8.4
Movement, pause, L3 and R3	The left stick moves the fighter, the Menu button pauses, and L3 and R3 are unused in v1.0	7.4, 7.10
Head modifier	Holding RB takes any kick to the head, for that target’s point value	Section 9
Audio asset list	Every voice line, sound effect, ambience and music cue is planned by milestone, adding fighter voices, coach shouts and the referee’s side and winner calls	1.10, 8.5

19 Prototypes

No prototypes exist yet. Each one links here when built.

Prototype	Question it answers	Contents	Milestone
Capture feasibility test	Which capture method records fast kicks and spins accurately?	The same back-leg roundhouse kick and spin hook kick through each candidate method and solver; compare supporting-foot pivot, chamber, hip rotation, trajectory, rechamber and landing	M1
Proof of concept	Are the Skill Stick and lead/back-leg model technically feasible?	Greybox arena, mannequin fighters, idle and movement, side kick, roundhouse kick, spin side kick, Skill Stick and modifier recognition, stance-based leg resolution, attack phases, foot traces for head, body or miss, basic scoring and stamina, one retargeted capture	M1
Combat prototype	Does the system create real sparring decisions: faking, countering, distance, body or head?	5–6 techniques, local two-player combat, stamina, scoring, rounds, active defense, switching sides, guards, combat camera, debug overlay with input icons	M2

20 Development Schedule and Milestones

The capstone runs to five milestones across two halves. The First Half is COMP 8800 Major Project 1, from 7 September to 7 December 2026: design, prototyping, Milestones 1 and 2 and four GDD drafts, ending with playable demo gameplay. Its dates follow the COMP 8800 course schedule. The Second Half (January to spring 2027) covers Milestones 3 to 5, Career and the gold build; its dates are added when BCIT publishes the 2027 schedule (18.1).

Period	Phase	Deliverables
Weeks 1–3: Sept 7-21 2026	Design	High-level concept (week 1), product design (week 2) and game systems design (week 3). GDD Draft 1 due 21 September
Weeks 4–6: Sept 28 – Oct 12 2026	Prototype	Prototyping toward Milestone 1. Draft 1 presented 28 September, with instructor and second-reader feedback
Weeks 7–10: Oct 19 – Nov 9 2026	Milestone 1	Skill Stick and motion capture proof of concept (section 19). GDD Draft 2 due 19 October, presented 26 October, feedback 2 November. Milestone 1 due 9 November
Weeks 11–13: Nov 16 – 30 2026	Milestone 2	Playable two-player combat prototype: the demo gameplay that closes the First Half (section 19). Milestone 1 presented, with feedback, 16 November. GDD Draft 3 due 23 November; Final Draft due 30 November
Week 14: Dec 7 2026	Presentation week	Milestone 2 due and presented, with feedback on Milestone 2 and the final GDD draft
December 2026 break	Buffer	Capture preparation, mocap production and cleanup
Winter 2027	Milestone 3	Full combat and animation system with the production mocap library
Winter 2027	Milestone 4	Complete game flow, AI opponent, local Tournament and the Career season: calendar, training, grading, tournaments and rank-gated technique unlocks
Spring 2027	Milestone 5	Beta testing, balancing, polish and requirement verification
Spring 2027	Final	Gold build, working demo and final report

20.1 Milestone Success Criteria

Milestone	Success criterion
GDD drafts	Draft 1 (21 September), Draft 2 (19 October), Draft 3 (23 November) and the Final Draft (30 November) submitted on time, with instructor and second-reader feedback addressed
M1 · due Nov 9 2026	Two fighters execute techniques via the Skill Stick; one capture proves the pipeline

Milestone	Success criterion
M2 · due Dec 7 2026	Two local players complete a basic sparring match, playable as a demo. If Milestone 1 cannot capture clean spinning kicks, this build is centred on the grounded kicks and whatever the lab session captured. Spinning and jump kicks stay in the move list with a temporary placeholder, the grounded technique with a boost in height, rather than being cut (16.4).
M3	Planned combat playable with natural transitions
M4	Startup to finished match against a local player or the AI; a Career season playable from the first club night to a tier 3 final
M5	Requirements verified; ready for demo

Epics per milestone: 8.13. Beyond v1.0: section 17.

21 Change Log

Date	Version	Author	Changes
September 2026	0.1	Joey Driedger	Preliminary project plan.
2026-09-12	0.2	Joey Driedger	Earlier draft, built on an older controller layout.
2026-09-14	0.3	Joey Driedger	Full rewrite, with controller icons and a layout diagram. Adds the main page, player objectives, game world, user interface, game objects, localization, tools, ideas and expansions, prototypes, schedule and change log; starting values for stamina, movement and hit reactions; unresolved questions with a resolved list.
2026-09-19	0.4	Joey Driedger	Two phases in place of three: v1.0 is what the capstone delivers and Future is everything after it, so only deferred items carry a tag. Online play, online tournaments, video review, challenges, replays and ragdoll blending become Future work; local Tournament against AI opponents stays in v1.0. Retargets the game to console, with Windows as the development platform. Brings Career into the Second Half, with tournament formats, event preparation and rank-gated technique unlocks. Cuts weight classes and gender divisions. Updates scoring to current WT values: a spinning kick to the head scores 6, and spin hook and tornado kicks reach the body as well as the head. Records that holding RB takes any technique to the head, and that the motion capture method is still undecided. Adds a glossary (section 22), controller notation beside the icons in section 9, and console and Career entries in the risk register. Makes the document self-contained and corrects the scope, rank, scoreboard and cross-reference errors listed in 18.2.
2026-09-21	0.5	Joey Driedger	Adds Referee Mode with its own controls and a referee rating (3.5, 3.5.1); body types with height and weight, fighting styles, an overall rating and an OVR 99 ceiling for every Career fighter (5.4.1 to 5.4.3). Fixes contradictions: multi-season Careers are part of v1.0, holding RB takes kicks but not punches to the head, punches register on the trunk protector, rivals switch style only between seasons, and the podium moment moves to the world series final. Gives every Career tournament tier its own venue (6.2). A second pass adds the knockdown count (8.5, 11.2), head values for side kicks (9.1), Referee Mode in the main menu (7.5) and the missing Future items (17); a third pass corrects a cross-reference in 1.8 and wording in 1.6 and 6.2. The release is now written v1.0. Section 20 follows the COMP 8800 course schedule: four GDD drafts, Milestone 1 due 9 November and Milestone 2 due 7 December 2026.

Date	Version	Author	Changes
2026-09-22	0.6	Joey Driedger	Adds PlayStation trophies and Xbox achievements: one list of 35 with a Platinum and 1,000 Gamerscore, most named after pop culture and the ATA's belt meanings, with rules that keep every trophy earnable offline in v1.0 and free of farming, and how they are built (5.7). Names six easter eggs, from the Konami Code to Bullet Time (6.5). Adds trophies to the platform table (1.11) and the section 5 feature list, and opens certification and legal-check questions (18.1).
2026-09-28	0.7	Joey Driedger	Updates Current Status (1.2) to 2026-09-28: GDD Draft 1 presented, with instructor and second-reader feedback due 2 November; binary game assets (Content/) migrated from git to Perforce; a MotionBuilder test character prepared and the preliminary lab appointment completed for the BCIT Vicon Nexus motion-capture session, with the capture session itself still to come.

Date	Version	Author	Changes
2026-09-29	0.8	Joey Driedger	Resolves four open questions raised in Draft 1 presentation feedback (18.1). Defines Active Defense as a Skill Stick defense mode (8.8). Resolves the roundhouse/hook direction: RS toward the side of the leg throwing the kick is the roundhouse, the other side is the hook, mirrored when RT selects the back leg since it's on the opposite side from the lead leg (7.2, 9.1, 18.2). Fixes the Tornado Kick's conflict with Fake Attack by moving its windup off LB entirely: it taps Y like every other jump kick, and its own full circle, starting downward, is what makes it the hardest input in the game (7.2, 9.2, 18.2). Builds out the jump system that 5.5 and 8.9 already implied: tapping Y just before a side, roundhouse, hook, axe or spin hook kick's gesture throws it airborne instead of grounded, with five new Jump kick rows in the move list (9.1, 9.2), the Red-rank unlock reworded to name them (5.5), and an animation-reuse mitigation so jump kicks share their grounded kick's capture rather than needing their own (16.4). Adds a standing-versus-sliding rule: a neutral left stick keeps a technique rooted in place, holding a direction slides or steps the fighter that way during it (7.4). Makes every gam-jeom penalty explicit and implementable — concrete thresholds for the boundary, falling, avoiding/delaying, leg lifting, below-the-waist and grabbing fouls, and a 5-gam-jeom round-loss limit (8.4) — and spells out exactly how a real knockdown is told apart from a Falling foul via which combat state transition triggers Knocked Down (8.10, 16.3). Reframes the platform target: the First Half milestones (M1, M2) demo on Windows only, with PlayStation 5 and Xbox Series X|S staying the v1.0 target for the finished capstone in Spring 2027 (1.11, 16.4, 18.2). Adds a placeholder-animation fallback for spinning kicks if Milestone 1 capture is not clean by Milestone 2 (16.4, 20.1). Updates the controller-layout diagram to match (7.1). Fixes the avoiding/delaying penalty so Low Guard is never read as crouching — it's a normal guard choice (8.4). Adds 7.10 Controls Menu: an exhaustive, tab-by-tab text reference of every input in the game, for the pause menu's future Controls screen. Redesigns Active Defense's body coverage: holding LT alone now blocks punches and body kicks by closing the guard, with RS Down kept as an explicit leg check that risks the leg-lifting penalty if held too long (8.4, 8.8, 18.2, 7.10). Adds Dash-into-clinch: closing the gap with a Dash into contact range starts a clinch automatically, the same as both fighters pressing forward (6.4).

Date	Version	Author	Changes
2026-10-02	0.9	Joey Driedger	Console menu design. A hub-style main menu with LB and RB tabs; Sparring and Tournament use premade men's and women's fighters, and the created fighter and fighter creator belong to Career only (7.5, 10). A full Controls screen (7.10) with a controller diagram, a Fakes tab, stick icons for quarter and half circles, Dash / Clinch, and no Menus & System tab. The title screen shows only the game name and a start prompt. A clickable prototype and a Lucid wireframe sheet now exist (18.1).
2026-10-04	0.10	Joey Driedger	Audio. Fighters kihap, grunt when hit and breathe harder below 25 stamina, with at least two men's and two women's voice sets; coaches shout short English cues from the corner chairs; each referee has their own voice; crowds never boo; and a quiet pulse marks a round's last 10 seconds (1.10, 4.3, 8.9, 11.2). The referee names the side with a gam-jeom and calls the winner, with a time-out call for video review and injuries in Future (8.5, 11.2). Adds a fighter and coach voice volume (7.8), the Content/Audio folder and the SW_ and MS_ asset prefixes (16.5), and voice casting and the new referee calls to the open questions; the audio asset list is resolved and the art asset list stays open (1.2, 18.1, 18.2). Glossary: kihap, kye-shi and seung (22).
2026-10-16	0.11	Joey Driedger	Feedback pass. Marks every [Future] section as non-essential to the first v1.0 release, documented in case the main features finish ahead of schedule (1, 17). Confirms PlayStation 5 and Xbox Series X|S as the v1.0 target, with Windows as the development and demo platform only (1.11, 16.4). Sharpens the December fallback if motion capture of spinning kicks fails: Milestone 2 centres on the grounded kicks and whatever the lab session captured, with the grounded technique plus a boost in height as the temporary placeholder for spinning and jump kicks (16.4, 20.1).
2026-10-30	0.12	Joey Driedger	Control and scope decisions. Kicks without RT use the lead leg, the spin hook requires RT, the left stick moves the fighter, the Menu button pauses and L3 and R3 are unused (7.4, 7.10). Defines the four guard postures (8.7). The Men or Women choice picks the roster only, and either roster can face the other (7.5, 10.1). Records the open conflicts found checking the 2026 WT rules (18.1).

22 Glossary

Taekwondo, competition and engine terms as this document uses them. Korean terms are romanized, with the hangul where the sport writes it.

Term	Meaning
Active Defense	A held defensive state on LT that reads the Skill Stick as defense: held alone it blocks punches and body kicks, RS Up blocks high, RS Down is a leg check, RS Left/Right sways or steps, instead of a kick (8.8).
ATA	ATA Martial Arts, formerly the American Taekwondo Association. Three trophy names come from the ATA, two of them from the meanings of its Songahm belts (5.7).
charyeot (차렷)	Referee command: attention, called before the bow (8.5).
Chung (청)	The blue side: blue protector, helmet and corner. The match assigns it; it is never a player's choice (10.2).
closed stance	Both fighters lead with the same leg.
dan (단)	Black-belt grade, counting up from 1st dan (5.3).
dobok	The taekwondo uniform: jacket, trousers and belt.
dojang (도장)	Training hall. The game's Training mode is set in one (3.2).
Equal Terms	The option that sets every attribute to 70, so skill alone decides the match (5.6).
gam-jeom (감점)	A penalty. It awards one point to the opponent (8.4).
Gamerscore	Xbox's points for achievements. Open Stance's list is worth 1,000 (5.7).
gup (급)	Coloured-belt grade below black belt, counting down from 10th to 1st (5.3).
gyeorugi (겨루기)	Sparring: the competitive, full-contact form of taekwondo, and the only form this game covers (1.1).
hogu	The trunk protector. In competition it is electronic and registers scoring kicks and punches (11.3).
Hong (홍)	The red side: red protector, helmet and corner (10.2).
joon-bi (준비)	Referee command: ready (8.5).
kal-yeo (갈려)	Referee command: break, separating the fighters (8.5).
keu-man (그만)	Referee command: stop, ending the round (8.5).
kihap (기합)	A fighter's shout when striking (1.10).
kye-shi (계시)	Referee command: time-out, stopping the clock for a video review or injury **[Future]** (8.5).
kye-sok (계속)	Referee command: continue (8.5).
kyeongrye (경례)	Referee command: bow (8.5).
Leader Pose	The Unreal Engine method that makes each gear piece follow the body's skeleton (16.1).

Term	Meaning
MetaHuman	Unreal Engine's photoreal human character system, used for every fighter and referee (16.1).
MoveData	The data table holding each technique's timing, stamina cost, points, target and range (16.3).
open stance	The two fighters lead with opposite legs, so their trunks face each other and the roundhouse kick is open. The game takes its name from it (1.1).
OVR	Overall rating: the average of a fighter's eight attributes. Every Career fighter can reach 99 (5.4.1).
poom (품)	Youth black-belt grade, held under 15, with a red-and-black belt (5.3).
PSS	Protector and Scoring System: the electronic protectors and scoreboard that register hits (8.3).
seung (승)	Referee call: wins, said after the winner's side, as in Hong seung (8.5).
shi-jak (시작)	Referee command: start, and the clock runs (8.5).
Skill Stick	This game's name for the right stick, which chooses the kick (8.6).
trophy	PlayStation's name for an achievement, graded Bronze, Silver, Gold or Platinum. Open Stance's trophies and Xbox achievements share one list (5.7).
WT	World Taekwondo, the sport's international federation, whose competition rules the game follows (8.2).